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Identification of Synergies and Causal Relationships among Sustainable Development Goals at the Local Level in Germany: An Empirical Study Based on Machine Learning

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Abstract

The United Nations 2030 Agenda for Sustainable Development emphasizes the importance of advancing goals in a synergistic manner. However, at the local level—particularly at the municipal scale, the interlinkages among Sustainable Development Goals (SDGs) remain insufficiently identified and quantitatively modeled. This study develops a machine learning framework that integrates network analysis, clustering, supervised regression, and causal inference, using publicly available SDG indicator data from German municipalities. The framework aims to systematically analyze the synergies, spatial heterogeneity, and causal pathways among local-level SDG targets.

The research begins by constructing an SDG goal network based on Spearman rank correlations and identifies synergistic communities through the Louvain algorithm. K-means clustering is then employed to categorize municipalities into distinct SDG development types. A random forest model is used to uncover key driving factors behind individual goals, while the Isolation Forest algorithm detects municipalities with anomalous performance. Finally, causal pathways with time-lagged effects among SDGs are identified through structure learning and panel data regression.

The results reveal that goals related to education, gender equality, and urban sustainability occupy central positions within the SDG network. Some municipalities exhibit significant deviations across multiple SDG dimensions, suggesting the need for tailored governance strategies. Moreover, the study finds that indicators such as the “gender income gap” have a significant impact on green development metrics like “electric vehicle adoption,” highlighting the critical role of social structures in sustainable transitions. This research offers a structured analytical toolkit and policy-oriented framework to support local SDG implementation.

Keywords: Sustainable Development Goals; Learning; Network Analysis; Causal Inference; German Municipalities

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1. Introduction

1.1 The SDGs as an Interconnected System

Since the United Nations adopted the 2030 Agenda for Sustainable Development in 2015, global efforts toward sustainability have entered a new era of integrated and strategic governance. The 17 Sustainable Development Goals (SDGs) span multiple domains—poverty reduction, education, health, energy, climate action—and form a globally recognized framework for achieving a more equitable, green, and inclusive future.

Importantly, the SDGs do not operate in isolation. They constitute a complex system where goals can reinforce or constrain one another. For instance, infrastructure investment (SDG 9) may promote economic growth and employment (SDG 8), while simultaneously increasing environmental pressure and potentially undermining climate objectives (SDG 13). Understanding these interdependencies—both synergies and trade-offs—is essential for integrated policymaking, especially at the local level where resources are limited and governance is often fragmented.

1.2 The Challenge of Local SDG Implementation

While the 2030 Agenda for Sustainable Development highlights the interconnected nature of the 17 SDGs, most empirical research has remained at national or global scales, relying heavily on expert judgment, aggregated indices, or qualitative semantic networks. These approaches, while informative, face three main limitations when applied to local governance contexts: First, there is the problem of data granularity and heterogeneity. Local governments usually report disaggregated SDG indicators, but these vary greatly in availability, quality, and coverage across time. As a result, conventional frameworks struggle to cope with such fragmented datasets, often producing inconsistent assessments of interlinkages. Second, advanced analytical tools are still underused. Many local SDG assessments rely on static reports, which makes it difficult to capture how synergies and trade-offs evolve over time. This lack of dynamic, data-driven approaches leaves important interactions hidden. Third, interpretability for policy is limited. Current analyses often stop at correlation matrices or expert scoring exercises. These may highlight some relationships but provide little guidance on how local policymakers should prioritize goals or anticipate the ripple effects of their interventions.

Germany provides an ideal testing ground to address these challenges. As a frontrunner in SDG localization, many German municipalities conduct Voluntary Local Reviews (VLRs) and publish subnational indicator data across multiple domains—education, health, mobility, equity, and environmental protection. However, these data are rarely exploited to uncover the underlying network structure and causal pathways that shape sustainable transitions at the local scale.

1.3 Research Objectives and Analytical Framework

This study aims to narrow the current gap by proposing a machine learning-based analytical framework that can identify synergies, trade-offs, and causal dependencies among SDG indicators at the municipal level in Germany. The framework addresses the following research questions:

- Are there stable and interpretable patterns of interlinkages among local SDG

indicators, and how do these patterns change over time?

- Can municipalities be classified into distinct development profiles based on their multi-SDG performance patterns?
- Which indicators act as key drivers or “leverage points” that strongly influence other SDG outcomes?
- Are there detectable causal pathways or time-lagged effects linking social, economic, and environmental goals at the local level?

To address these questions, the study integrates network analysis, unsupervised clustering, supervised regression, anomaly detection, and causal inference into a unified pipeline. This approach enables a transition from simple correlation mapping to structural understanding and causal interpretation, offering actionable insights for local policymakers.

Following the introduction, the remainder of the thesis is structured as follows: Chapter 2 reviews relevant literature on SDG interlinkages; Chapter 3 introduces the data and methodology; Chapter 4 presents empirical results; and Chapter 5 concludes with a summary of findings and future research directions.

2. Literature Review

2.1 The Complexity of SDG Interlinkages and Local Challenges

The adoption of the 2030 Agenda for Sustainable Development by the United Nations in 2015 introduced 17 Sustainable Development Goals (SDGs) aimed at achieving a balance among economic, social, and environmental objectives (United Nations, 2015). However, the relationships among these goals are highly complex, involving both synergies and trade-offs that significantly influence implementation effectiveness (Horvath et al., 2022). According to Chaniotakis et al. (2024), "the interlinkages domain is still exploration, largely neglecting the local level and being disconnected to policy making." These interlinkages are particularly underexplored at the local level, where municipalities, such as those in Germany conducting Voluntary Local Reviews (VLRs), often lack systematic tools to assess the interactions among SDGs (Meuleman, 2021).

Early studies on SDG interlinkages primarily relied on expert judgment, qualitative scoring, or matrix-based methods to identify relationships, which were often subjective and not well suited for cross-case comparison. More recent approaches have incorporated network analysis, semantic mining, and machine learning to build interlinkage models. For instance, Horvath et al. (2022) reviewed 30 methods for quantifying interlinkages, highlighting the fragmentation of existing approaches and calling for a more systematic comparative framework, including correlation-based networks and semantic analysis.

In the area of semantic analysis, Song and Jang (2023) used Word2Vec to construct a semantic network of SDG targets, moving beyond traditional indicator correlations. Their study revealed that inter-goal relationships often cut across traditional goal boundaries, suggesting that policy design should target cross-sectoral communities rather than rigid SDG classifications. However, these innovative approaches have yet to be widely applied at the local level in contexts such as Germany.

2.2 Applications for Machine Learning in SDG Analysis

Machine learning has increasingly been used in SDG-related empirical studies. The *SustainBench* platform provides and disposes benchmark datasets across seven SDG domains—such as health, education, and climate—to evaluate the performance of predictive models using machine learning (Yeh et al., 2021). Asadikia et al. (2021) applied Boosted Regression Trees to identify effective SDG combinations across more than 150 countries, determining which goals most strongly drive overall SDG Index improvements between 2017 and 2019.

Vinuesa et al. (2019) conducted a systematic assessment of the potential positive and negative impacts of artificial intelligence (AI) on SDG achievement. They found that AI could accelerate progress toward 134 targets (around 79%) by improving efficiency and precision in education, healthcare, environmental management, and energy systems. However, AI also poses risks to 59 targets (about 35%), including widening social and economic inequalities, data privacy violations, algorithmic bias, and increased carbon emissions. The authors emphasized the need for robust ethical frameworks and governance mechanisms to ensure transparency, fairness, and accountability in AI deployment, particularly in developing countries.

2.3 Localized SDG Monitoring: Insights from VLR Practices

Several scholars have examined how cities engage in SDG monitoring through VLRs. Meuleman (2021) and Ciambra et al. (2023) found that local governments retain significant autonomy in choosing and reporting SDG indicators, with choices influenced by local administrative traditions, data availability, and policy priorities. In cities like Cologne and Düsseldorf, the focus is often placed on SDGs closely aligned with municipal mandates, such as SDG 11 (Sustainable Cities and Communities) and SDG 4 (Quality Education), while goals requiring cross-sectoral or international coordination—such as SDG 15 (Life on Land) and SDG 17 (Partnerships for the Goals)—receive less attention.

Ciambra et al. (2023) introduced the *Indicator Proximity Index (IPI)* to evaluate the alignment between local indicators and the standardized VLR framework across 29 European cities, analyzing 2,354 indicators in total. Their findings showed that the selection of indicators is shaped more by local data accessibility and policy relevance than by administrative size or governance type. Cities tend to prioritize measurable indicators in education, health, and infrastructure, while monitoring capacity remains limited for ecological and global partnership goals.

While these studies provide valuable insights into the logic behind indicator selection, their methodologies have largely relied on qualitative assessments, text clustering, or semantic matching. A significant gap remains in using machine learning to explore interlinkages at the local level, particularly in building structured SDG networks, conducting cluster analyses, or identifying causal relationships—which limit the potential to design more predictive and responsive governance systems.

2.4 Toward a Data-Driven Framework for Local SDG Interlinkage Modeling

A range of advanced analytical tools have been proposed to capture interlinkages and local dynamics more systematically. Spearman correlation networks are commonly used to represent nonlinear rank-based relationships among SDG indicators, and Louvain algorithms can detect community structures within these networks (Wang et al., 2023). Compared to semantic methods, this approach is more suitable for processing quantitative municipal-level data and capturing real-world interaction patterns (Laumann, von Kügelgen & Barahona, 2020).

Spatially, clustering methods such as K-means have been applied to group regions based on SDG performance. For example, Castelli, Mocenni, and Dimitri (2024) identified geographic heterogeneity in SDG 2, 12, and 13 implementations across Italian regions. Applying similar methods in Germany could reveal how interlinkage patterns vary spatially, providing a basis for region-specific interventions.

Regression tree algorithms such as Random Forest, XGBoost, and Boosted Regression Trees are effective tools for identifying key drivers of SDG outcomes due to their ability to model non-linear and interaction effects (Vinuesa et al., 2020; Asadikia et al., 2021). These models can highlight which indicators most strongly influence others, offering insights into the structural mechanisms behind synergies and trade-offs.

Anomaly detection algorithms, such as the Isolation Forest, can also help identify "outlier

cities" whose SDG profiles deviate significantly from normative patterns, offering early warning signals for policymakers (Wang et al., 2023). In recent years, structure learning approaches—such as Bayesian networks and the PC algorithm—have been used to infer directed causal pathways among SDG indicators, with panel OLS or VAR methods applied to verify statistical significance (Laumann et al., 2020). These causal models represent the frontier of SDG interlinkage research, offering deeper explanatory and predictive power than traditional correlation-based approaches.

Building on these advances, this study applies a multi-method machine learning framework to German municipal SDG data, combining Spearman-based network construction, community detection, spatial clustering, feature importance analysis, anomaly detection, and causal modeling. This integrated approach addresses three key gaps in the literature: (1) the lack of local-scale interlinkage analysis; (2) the need for higher-precision, interpretable models; and (3) the absence of visualization and diagnostic tools for policy application. The resulting framework can help local governments identify goal clusters, prioritize policy areas, and optimize resource allocation. It also enables cities to learn from similar peers, target structural drivers, and course-correct anomalous patterns in real time.

3. Methodology

This study develops a multi-stage machine learning framework to identify, classify, and interpret interlinkages among Sustainable Development Goals (SDGs) at the municipal level in Germany. The framework is designed as an integrated pipeline that progresses from data cleaning to structural mapping, pattern discovery, driver identification, and finally causal inference, ensuring that each step builds upon the previous one to derive increasingly interpretable insights. The methodology consists of five interconnected components:

1. The raw municipal-level SDG indicators were first cleaned and imputed to deal with missing values and then standardized to improve comparability across time and across regions. This stage creates a reliable basis for subsequent network analysis and modeling.
2. Using Spearman rank correlations, undirected weighted networks were constructed for each year. Examining network metrics and comparing them over time helps to capture the underlying synergy structures and how they evolve, which then provides a reference point for community detection.
3. To explore goal-level groupings, the Louvain algorithm was applied to identify internally cohesive clusters of indicators. At the same time, municipalities were grouped into broader development profiles through K-means clustering. Together, these two methods shed light on both the interdependencies among goals and the heterogeneity across municipalities.
4. Random Forest models were then used to examine directional associations, focusing on which indicators tend to have stronger predictive power for others. This analysis draws attention to potential leverage points where interventions could have wider system-level consequences.
5. Anomaly Detection and Causal Inference –Finally, Isolation Forest algorithms were employed to flag municipalities with unusual SDG patterns, offering cases for closer inspection. Building on this, structure learning and panel regression techniques were applied to investigate time-lagged causal pathways among SDG indicators, going beyond simple correlations and pointing toward possible mechanisms of influence.

By linking these modules within a single analytical pipeline, the study avoids having results that are separated by method and instead creates a process of progressive refinement. In practice, correlation analysis can point to potential connections, clustering helps to uncover broader structural patterns, regression brings attention to influential drivers, and causal analysis allows these relationships to be tested for directionality. Taken together, this step-by-step approach makes the results easier to interpret and more robust. More importantly, it provides a way for local policymakers to think about which goals should be prioritized, where synergies might be leveraged, and how cross-sectoral strategies could be designed.

3.1 Data Preprocessing

All data preprocessing, modeling, and analysis were conducted using Python 3.7. The raw data were obtained from publicly available SDG indicator datasets published by German municipalities. These datasets, in CSV format, included fields such as "municipality", "state", "SDG", "indicator name", "indicator description", "trend", data source (URL), and JSON-

formatted time series values.

The JSON content was first parsed and transformed into structured Pandas DataFrames, generating both long and wide table formats suitable for subsequent time series and panel data analysis. Using the pandas library and ydata_profiling.ProfileReport, we examined the basic characteristics of the data—size, types, value ranges, and missing values. For missing data, a hierarchical imputation strategy was applied: if an indicator had more than 80% missing values, it was excluded; otherwise, linear interpolation (interpolate(method='linear')) was used to fill in gaps for time series with sufficient continuity. Data from 2019 to 2022 were retained.

To ensure comparability and robustness across indicators, all numerical variables were standardized using Z-score normalization:

$$z_j = \frac{x_i - \bar{x}}{\sigma} \tag{1}$$

where x_i is the raw value, $\bar{x}$ is the mean, and σ is the standard deviation of the indicator. This step eliminates unit inconsistencies and enhances the precision of correlation and distance-based algorithms.

3.2 Spearman Correlation Network Construction

"Spearman's rank correlation coefficient is a nonparametric measure of monotonic association and is robust to non-normal distributions and outliers" (Spearman, 1904; Spearman's rank correlation coefficient, 2025). In sustainability studies, Spearman correlation has been commonly used to assess interlinkages among SDGs, as it can "reveal structural relationships among SDG indicators where linear assumptions fail" (Laumann, von Kügelgen & Barahona, 2020).

To explore the structural evolution of SDG interactions at the local level, Spearman rank correlation networks were estimated at both the 70 indicators and 17 SDG goal levels for each year from 2019 to 2022. For the SDG-level analysis, standardized indicator values were averaged by municipality, SDG, and year to build a time-series matrix. The indicator-level matrix retained the original format of municipalities by indicators.

Spearman rank correlation coefficients were computed between all pairs of variables to capture their rank-based associations, which are robust to non-normality and outliers. The coefficient ρ is defined as:

$$\rho = 1 - \frac{6\Sigma_{i=1}^{n} d_i^2}{n(n^2 - 1)} \tag{2}$$

where d_i is the rank difference for each sample pair, and n is the number of observations. This method has stronger robustness against non-normal distributions and extreme values and can more accurately reflect the correlation strength among the various indicators of the method.

After constructing the full correlation matrix, weak correlations with $|\rho| \geq 0.3$ were set to zero to reduce noise. The filtered matrix was used to generate an undirected weighted graph using a spring layout. Additionally, heatmaps were plotted to visualize the strength of inter-indicator and inter-SDG correlations.

3.3 Dynamic Network Evolution and Robustness Testing

Network structural changes across years were analyzed using edge difference and Jaccard similarity metrics to quantify the number of new and removed edges and assess network stability. Degree centrality and betweenness centrality were calculated to measure node influence over time, with the top five nodes reported annually.

Partition similarity was further used to evaluate community stability over time. For each pair of adjacent years, Louvain-based community structures p_t and p_{t+1} were generated, and their similarity was measured using the Jaccard index:

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|} \tag{3}$$

and Normalized Mutual Information (NMI):

$$NMI(P_t, P_{t+1}) = \frac{2I(P_t; P_{t+1})}{H(P_t) + H(P_{t+1})} \tag{4}$$

where I denotes mutual information and H represents entropy.

To test robustness, networks were reconstructed under different correlation thresholds (0.2–0.4), and the sensitivity of network density and average clustering coefficient was assessed. Bootstrap resampling (n=100) with replacement was performed on the municipality sample to rebuild the merged network, and 95% confidence intervals were calculated for network density and the top five nodes' degree centrality. These procedures provide a comprehensive framework for assessing temporal dynamics and structural robustness in SDG correlation networks.

3.4 Community Detection

The Louvain algorithm remains a standard approach for detecting modular structures in large, weighted networks because it "provides high-quality partitions in large, weighted networks with computational efficiency" (Blondel et al., 2008; Fortunato & Hric, 2016). Communities refer to closely connected subsets of indicators that reflect coordinated SDG groupings at the local level. The Louvain algorithm was applied to the undirected weighted graph to detect communities. Modularity Q was calculated to evaluate the quality of the partition:

$$Q = \frac{1}{2m} \sum_{i,j} \left(w_{iJ} - \frac{k_i k_j}{2m} \right) \delta \left(c_{i_J} c_j \right) \tag{5}$$

Where w_{ij} is the edge weight between nodes i and j, k_i is the weighted degree of node i, m is half the total network weight, and $\delta \left(c_{i_J} c_j \right)$ indicates whether nodes belong to the same community.

Communities were filtered based on size (≥5 nodes) to focus on representative groupings. For each community, internal properties were quantified, including mean internal edge weight (i.e. mean Spearman correlation) and mean degree centrality, reflecting internal cohesion and structural tightness. Using a pre-constructed SDG–indicator mapping, the distribution of SDG nodes within each community was analyzed to support functional interpretation.

The most frequent 1–2 SDG nodes in each large community were translated to their English names to serve as community labels, enhancing interpretability and thematic clarity. The complete network structure was visualized using a spring layout.

3.5 Unsupervised Clustering

To identify performance patterns, the standardized SDG scores of each municipality from 2019 to 2022 were averaged to construct multi-dimensional feature vectors. Missing values in each SDG dimension were imputed using the median:

$$\bar{x}_{ij} = \begin{cases} x_{i_j}, & x_{i_j}\ non-missing \\ me\ dian(x_1 j, x_2 j, \ldots, x_n j,), & x_{i_j}\ \ missing \end{cases} \tag{6}$$

This approach is robust to outliers and preserves the central tendency of each distribution.

K-means is "one of the most commonly used methods for partitioning multivariate SDG data into homogeneous groups" (MacQueen, 1967; Jain, 2010). K-means clustering with fixed cluster number K=5 was applied to minimize within-cluster variance:

$$J = \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|^2 \tag{7}$$

Here, μ_k represents the centroid of the k-th cluster, and $\|x_i - \mu_k\|$denotes the Euclidean distance. By fixing the number of clusters K = 5 and conducting multiple iterations for optimization, similar regions can be grouped into the same cluster, thereby revealing the clustering patterns in multi-objective performance. After clustering, the size of each cluster is statistically analyzed, and the centroid positions are compared to characterize the strengths and weaknesses of different types of regions in terms of sustainable development.

To visually present the high-dimensional clustering results, principal component analysis is applied to map the feature vectors onto a two-dimensional plane. Specifically, through the eigen-decomposition of the covariance matrix $\Sigma = \frac{1}{n}\Sigma_i(x_i - \bar{x})(x_i - \bar{x})^T$, $\Sigma v_m = \lambda_m v_m$, the first two eigenvectors v_1 and v_2 corresponding to the largest two eigenvalues λ_1 and λ_2 are taken as the projection basis, obtaining the two-dimensional coordinates $[v_1, v_2]^T x_i$. The distribution of cluster points in the scatter plot can be used to evaluate the separability between clusters and the internal compactness. Finally, the coordinates of the cluster centers in the original 17-dimensional space are displayed, which helps to reveal the relative contribution of each cluster to each target. This module can not only identify the natural groupings between regions but also provides a basis for subsequent causal analysis.

3.6 Supervised Regression

To analyze potential interdependencies among Sustainable Development Goals (SDGs), ensemble decision tree regression models were employed to perform multi-target prediction and identify driving factors. For each target variable y, the remaining 16 SDG scores were used as predictors X to construct regression models. Five-fold cross-validation was conducted to evaluate prediction performance, with mean squared error (MSE) defined as:

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{8}$$

The mean and standard deviation of MSE were reported to reflect the relative predictability of each target.

Random Forest regression, introduced by Breiman (2001), is "particularly effective for exploratory modeling, as it provides internal estimates of variable importance while minimizing overfitting." Its ensemble learning approach has been used in recent SDG research to uncover nonlinear dependencies and predict goal performance (Breiman, 2001; Asadikia, Rajabifard & Kalantari, 2021). Based on the full training set, feature importance was extracted from the fitted random forest regressors. The importance score ϕ_j, for each feature was computed by aggregating impurity reductions at each tree split and normalizing across the forest. This score represents the relative contribution of the j-th predictor to the target. For each SDG, the top five most important predictors were identified to reveal the dominant association pathways.

A global feature importance matrix $\bar{\Phi} = [\phi_{ij}]$ was compiled, where rows and columns correspond to target and predictor goals, respectively. The matrix was visualized using a heatmap. In addition, bar charts were generated for each SDG to present its top five drivers. This supervised regression analysis quantified predictive relationships among SDGs and revealed potential structural dependencies, offering empirical guidance for causal inference and policy prioritization.

3.7 Anomaly Detection

Isolation Forest is efficient in high-dimensional sustainability datasets to identify municipalities with extreme or inconsistent SDG performance (Liu, Ting & Zhou, 2008; Wang et al., 2023). As Liu et al. (2008) describe, this algorithm isolates anomalies "based on the idea that outliers require fewer random splits to separate from the rest of the data". To detect municipalities that significantly deviate from typical SDG performance, an anomaly detection method based on Isolation Forests was used. This algorithm constructs a forest of $n = 200$ binary trees, where each tree randomly samples the feature space and recursively partitions it. The average path length $E[h(x)]$ for each observation reflects the number of splits required to isolate it.

The anomaly score for a given sample x is defined as:

$$s(x,n) = 2^{-\frac{E[h(x)]}{c(n)}} \tag{9}$$

$$c(n) = 2H(n-1) - \frac{2(n-1)}{n} \tag{10}$$

where $H(m) = \sum_{i=1}^{m} \frac{1}{i}$ is the harmonic number and $c(n)$ is a normalization constant. A score $s(x,n)$ close to 1 indicates a high likelihood of anomaly, whereas values close to 0 indicate normality.

The contamination rate $\epsilon = 0{,}05$ was set to flag the top 5% of samples with the highest anomaly scores. These outlier municipalities were then reviewed based on their SDG performance profiles to identify deviation patterns—such as overall underperformance, extreme values in single indicators, or inconsistencies across goals. This module enables the identification of structurally atypical cases and provides reference points for targeted interventions and case studies.

3.8 Causal Relationship Analysis

To uncover potential causal mechanisms among SDG indicators, a constraint-based structure learning algorithm was applied to construct a directed acyclic graph (DAG) representing directional dependencies. Conditional independence tests were conducted using partial correlation coefficients, defined as:

$$\rho_{XY|Z} = \frac{\rho_{XY} - \rho_{XZ}\rho_{YZ}}{\sqrt{(1-\rho_{XZ}^2)(1-\rho_{YZ}^2)}} \tag{11}$$

If the p-value of $\rho_{XY|Z}$ exceeded 0.05, the link between variables X and Y was considered conditionally independent given Z, and the corresponding edge was removed. This procedure was repeated iteratively until all statistically insignificant edges were excluded, resulting in a DAG where nodes represent SDG indicators and edges represent potential causal pathways.

Each directed edge $X \rightarrow Y$ in the DAG was treated as a causal hypothesis and tested using a panel regression model with lagged predictors:

$$Y_{i,t} = \alpha_i + \beta X_{i,t-1} + \varepsilon_{i,t} \tag{12}$$

where i indexes the municipality, t indicates the year, and α_i represents fixed effects. Ordinary least squares estimation was used with heteroskedasticity-robust standard errors. Only causal paths with $p < 0.05$ and consistent coefficient signs were retained. This framework enabled the identification of statistically valid, time-lagged causal relationships among SDG indicators, offering deeper insights into the mechanisms of sustainable development dynamics.

4. Results and Discussion

4.1 Data Preprocessing Results

First, all the indicators were extracted, totaling 70. The result is shown in table 1.

Table 1 Indicator 70

	SDG	indicator_name
0	1	SGB II / SGB XII rate
1	1	Poverty - child poverty
2	1	Poverty - youth poverty
3	1	Poverty - elderly poverty
4	1	Homelessness
5	2	Nitrogen surplus in agriculture
6	3	Premature mortality - women
7	3	Premature mortality - men
8	3	Basic supply close to home - general practitioner
9	3	Basic supply close to home - hospital
10	3	Basic supply close to home - pharmacy
11	3	Staff in nursing homes
12	3	Staff in nursing services
13	3	Nursing home places
14	3	Air pollution
15	4	Inclusive child day-care facilities
16	4	Child care (under 3s)
17	4	Basic supply close to home - primary school
18	4	School drop-out rate
19	5	Ratio of employment rates of women to men
20	5	Ratio of median income of women and men
21	5	Proportion of women in the city council, municipal council or district council
22	6	Wastewater treatment
23	7	Electricity from renewable sources
24	7	Charging point infrastructure
25	8	Gross domestic product
26	8	Long-term unemployment rate
27	8	Employment rate - 15 to 64-year-olds
28	8	Employment rate - 55 to 64-year-olds
29	8	Employed and receiving benefits (Aufstocker)
30	9	Broadband supply - private households
31	9	high qualified (%)
32	9	Start-ups
33	9	Highly qualified
34	10	Naturalisations
35	10	Employment rate - foreigners
36	10	School drop-out rate - foreigners
37	11	Land use intensity

38	11	Rental prices
39	11	Living space
40	11	Basic supply close to home - supermarket
41	11	Car density
42	11	Cars with electric motors
43	11	Traffic casualties
44	11	Land use
45	11	New land use
46	11	Local recreation areas
47	11	Completed residential buildings with renewable heating energy
48	12	Drinking water consumption - private households
49	12	Amount of waste
50	12	Fairtrade Town
51	12	Fairtrade Schools
52	13	Greenhouse gas savings through climate protection projects (t per inhabitant)
53	13	Greenhouse gas emissions - transport
54	13	Greenhouse gas emissions - municipal facilities
55	13	Settlement density in floodplain area
56	13	Greenhouse gas emissions - industry and manufacturing
57	13	Greenhouse gas emissions - private households
58	13	Greenhouse gas emissions - industry / commerce, trade, services and other
59	14	Watercourse quality
60	15	Nature conservation areas
61	15	Landscape quality
62	15	Non-dissected open space areas
63	16	Liquidity loans
64	16	Crimes
65	16	Financial resources balance
66	16	Tax revenue
67	16	Corruption Prevention Index (2022) (%)
68	17	Partnerships in Global South countries
69	17	Project partnerships with partners in countries of the Global South

The initial dataset contained 70 SDG-related indicators across German municipalities. However, missing data posed a significant challenge: more than 80% observations were missed before 2019, with over 10 indicators having data gaps exceeding 70% after 2019 (Figure 1, Figure 2). To improve data reliability, only 2019–2022 data was retained, highly incomplete indicators were excluded, and linear interpolation was applied to fill remaining gaps.

These results reflect broader issues in municipal-level SDG reporting, where fragmented and inconsistent data collection limits the ability to conduct long-term, comparable analyses (Ciambra et al., 2023). Environmental and global partnership indicators were particularly affected, which may bias the observed SDG interlinkages toward social and economic dimensions.

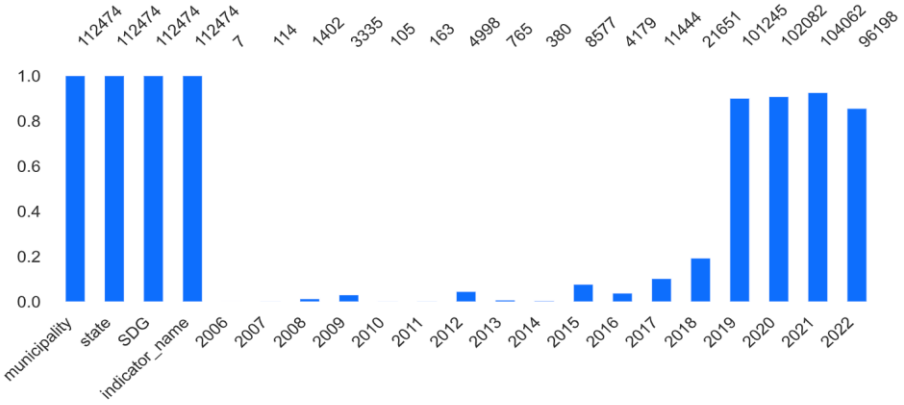


Figure 1 Overview of Missing Values

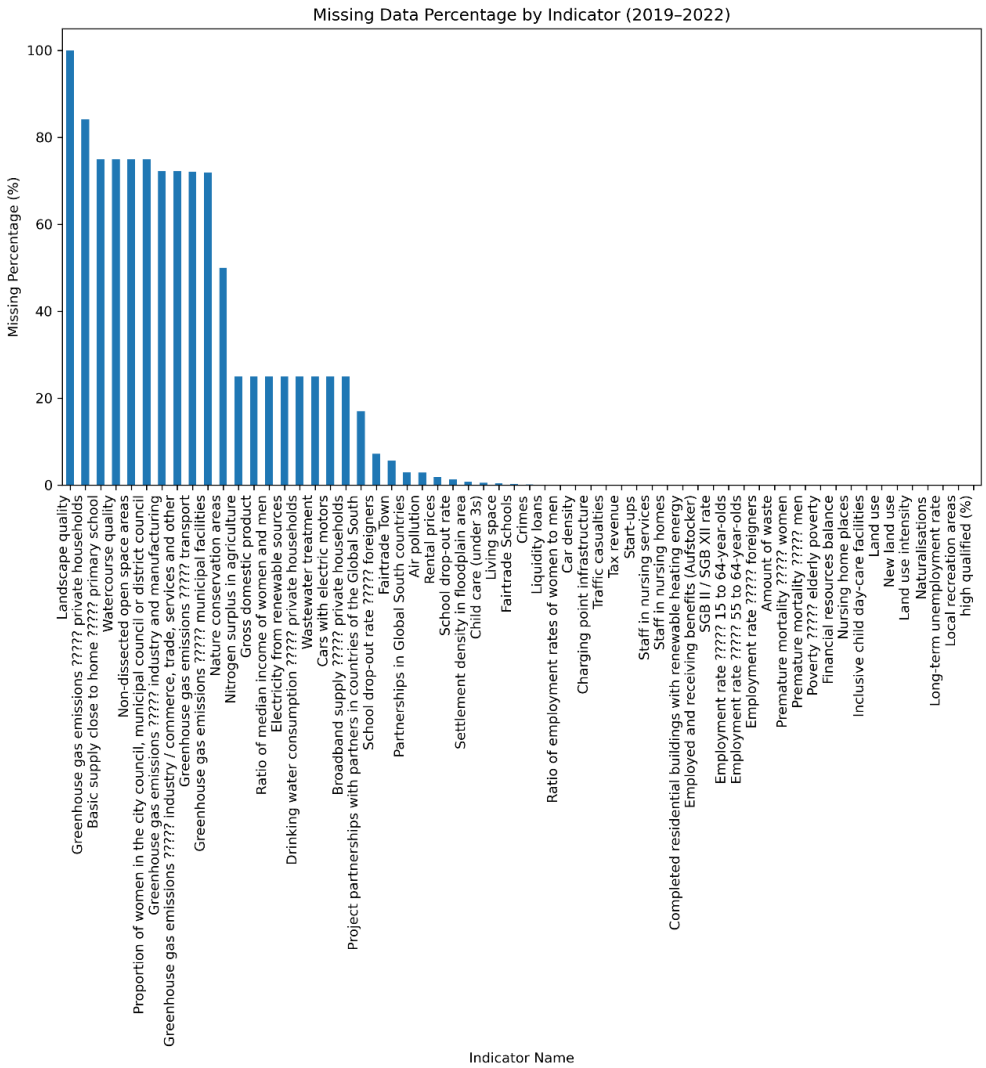


Figure 2 Missing Data Percentage by Indicator (2019-2022)

4.2 Spearman Correlation Networks

For indicator-level, the annual Spearman correlation heatmaps (Figure 3) show a consistently

dense core of social-economic indicators, with the strongest links observed among employment rate (SDG 8), land use intensity (SDG 11), and income-related indicators (SDG 1/10). These formed highly connected subgraphs with correlation coefficients frequently exceeding 0.6–0.7, indicating strong co-movements across municipalities. The 70 indicators spearman network graphs are included in the appendix. In 2021, the network density peaked, with an increased number of high-weight edges (>0.5), suggesting that local economic and social service provision became more synchronized, potentially due to post-pandemic policy responses.

Peripheral indicators such as renewable energy share (SDG 7) and drinking water consumption (SDG 12) had weaker connections (<0.3), remaining largely isolated from the main social-economic cluster. This fragmentation implies that resource-use indicators were not systematically integrated with broader development measures in most municipalities.

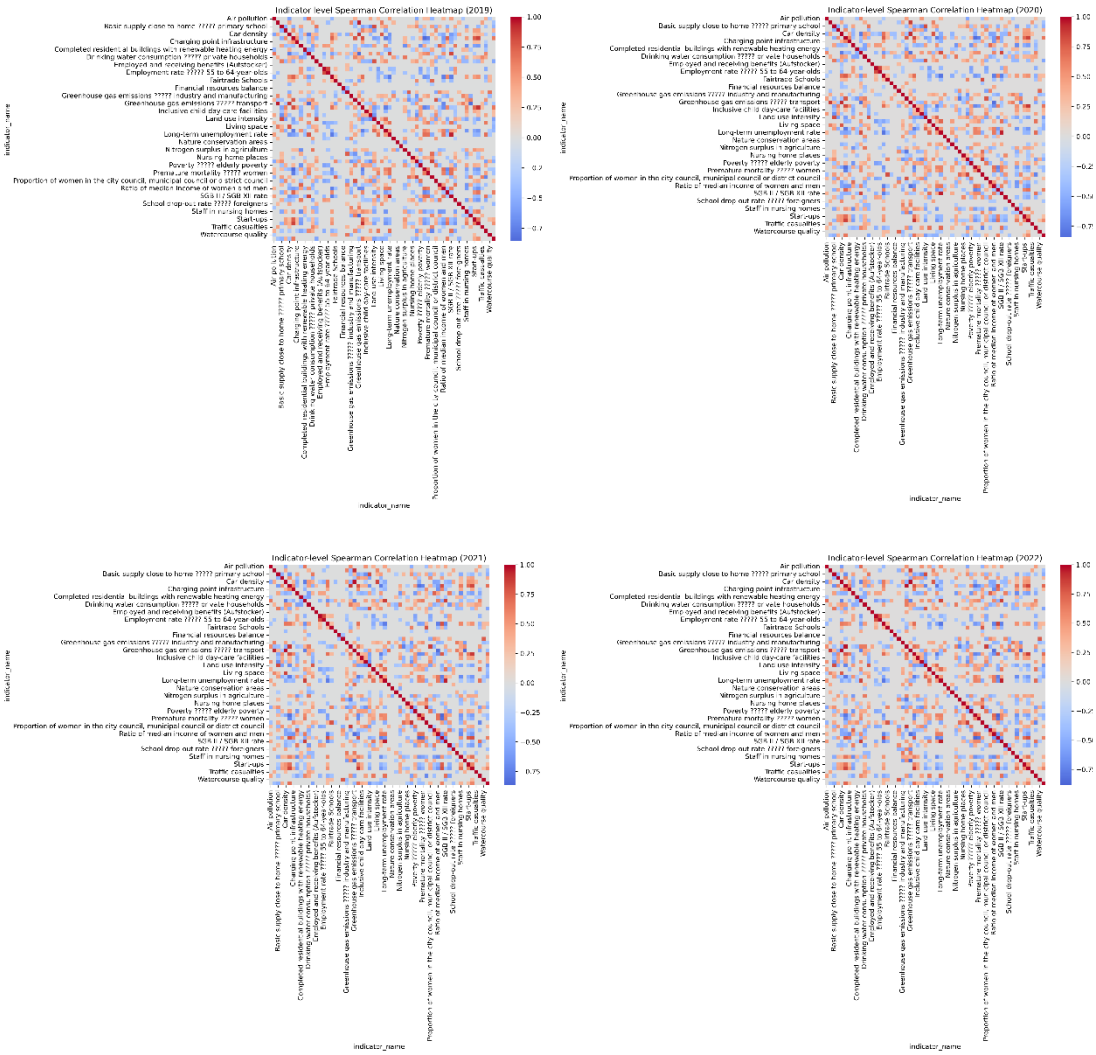


Figure 3 Indicator-Level Spearman Correlation Heatmap

For the SDG level, the 17 goals were treated as nodes, with edges defined by Spearman correlations between their respective average indicator scores (Figures 4). The 17 SDG goals spearman network graphs are included in the appendix. The network exhibited partial structural stability. Strong correlations were commonly observed between SDG 4 (Quality

Education), SDG 5 (Gender Equality), SDG 6 (Clean Water and Sanitation), and SDG 9 (Industry, Innovation and Infrastructure) with SDG 10 (Reduced Inequalities), suggesting these goals are often jointly addressed in local governance. In contrast, SDG 13 (Climate Action), SDG 15 (Life on Land), and SDG 8 (Decent Work and Economic Growth) showed relatively weaker associations with other goals, indicating that environmental and economic objectives may still be pursued independently at the municipal level.

For the SDG level, aggregating indicators by SDG (Figures 4) produced a less dense but stable network structure. The strongest correlations were observed between: SDG 4 (Education), SDG 5 (Gender Equality), and SDG 10 (Reduced Inequalities) ($\rho \approx 0.5$–0.6), reflecting close alignment between education access, gender inclusion, and equity-focused outcomes. SDG 6 (Clean Water) and SDG 9 (Industry & Innovation) also showed moderate links to SDG 10, suggesting infrastructure and resource provision co-evolve with social equity improvements. Conversely, SDG 13 (Climate Action) and SDG 15 (Life on Land) were weakly correlated with most other goals ($\rho < 0.2$), indicating that environmental objectives were often pursued independently of social-economic programs. SDG 8 (Decent Work) also had lower-than-expected connections, possibly due to heterogeneity in local labor market policies across municipalities.

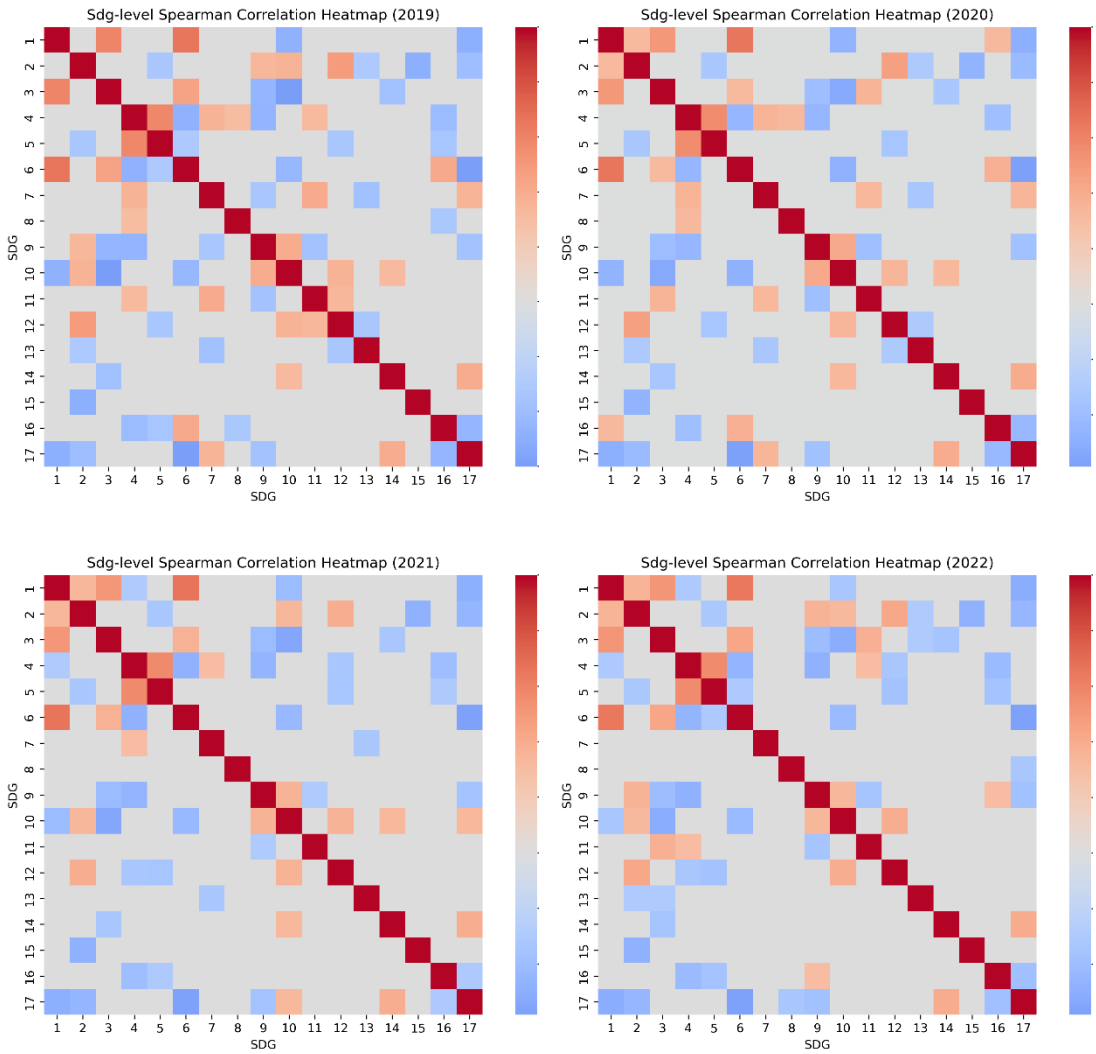


Figure 4 SDG-Level Spearman Correlation Heatmap

The correlation networks reveal that local SDG interactions in Germany are driven primarily by social and economic goals, forming a strong interlinked core, while environmental goals remain peripheral. The shift in 2021 toward higher connectivity suggests temporary policy coordination effects, but this was not sustained in 2022. This finding is consistent with Horvath et al. (2022), who also observed that social and economic goals tend to form a "core cluster" in SDG networks, while environmental goals remain relatively isolated.

4.3 Dynamic Network Evolution and Robustness Testing

To explore the evolution and structural stability of SDG relationships at the local level, correlation networks were constructed for each year at both the indicator and SDG levels, based on Spearman coefficients. The analysis focused on three dimensions: structural evolution, sensitivity analysis, and robustness testing.

The indicator-level correlation networks exhibited high structural volatility from 2019 to 2021. Edge turnover was substantial, with 65–69 new links added and 60–64 links removed each year (Figure 5). This suggests that local SDG interactions at the indicator level are dynamic, reflecting changing policy priorities or data variability across municipalities. However, between 2021 and 2022, new edge formation decreased to 43, signaling a possible structural convergence of relationships among indicators in the post-pandemic period. Jaccard similarity scores (Figure 6) confirmed this trend, rising from 0.795 (2019–2020) to 0.837 (2021–2022), indicating progressively more stable correlation structures over time. The observed stabilization reflects the gradual alignment of municipal policies or more consistent indicator reporting. Community structure similarity, measured by Normalized Mutual Information (NMI), reached its peak in 2020 (NMI = 0.632), suggesting that the strongest reorganization of indicator clusters occurred early in the study period. After 2020, the structure became more consistent, hinting at emerging policy synergies.

At the SDG aggregation level, network changes were more moderate but still significant. Edge additions exceeded removals only in 2021–2022, suggesting improved goal-level coordination in the most recent year of analysis. However, Jaccard similarity (Figure 6) dipped notably in 2020 (0.690), indicating temporary fragmentation of goal relationships, possibly due to disruptive events such as the COVID-19 crisis. By 2022, similarity recovered, pointing to partial re-integration of SDG policies across municipalities. Community stability (Figure 7) reached its highest NMI (0.741) in 2022, implying that goal-level clusters became more robust, with recurring groupings of social and economic goals co-occurring in municipal data.

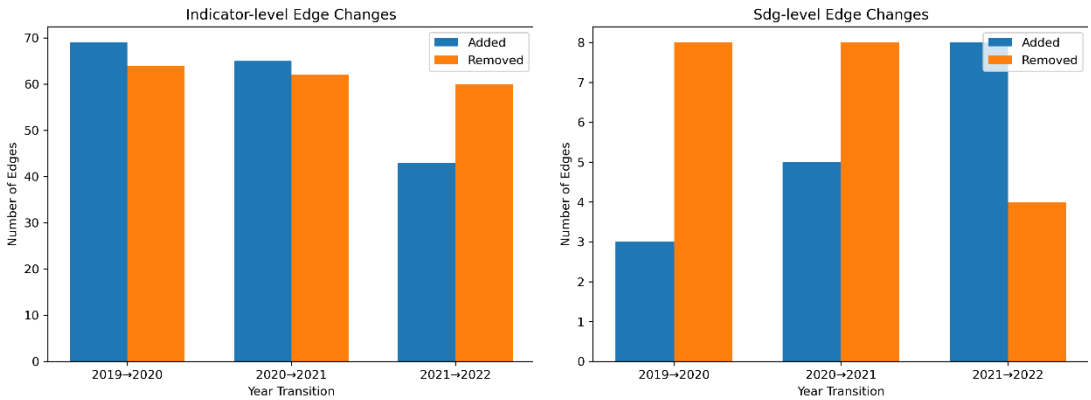


Figure 5 Edge Changes

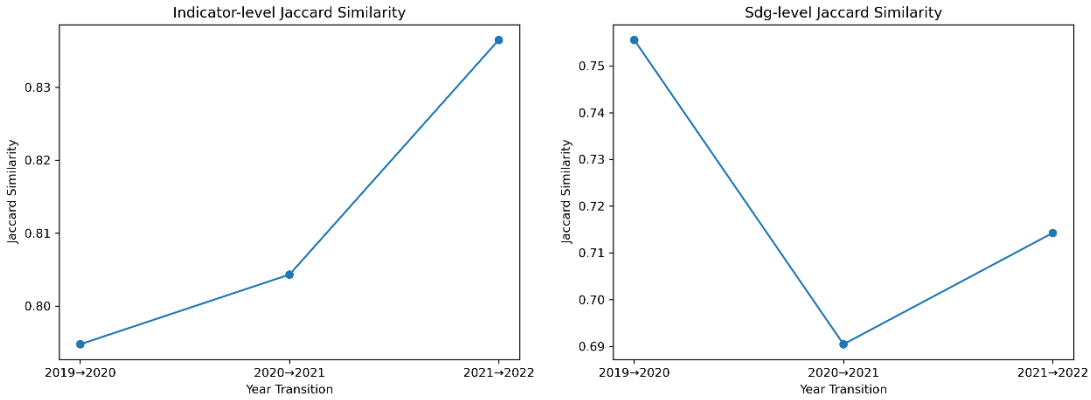


Figure 6 Jaccard Similarity

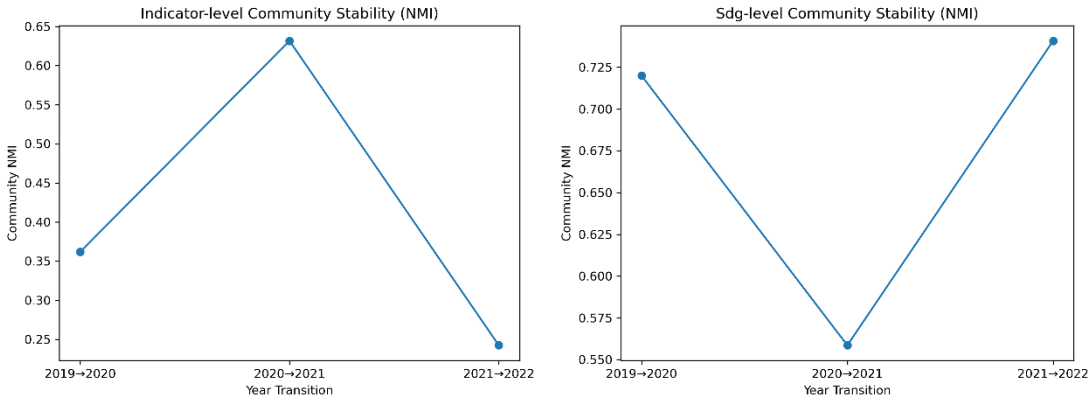


Figure 7 Normalized Mutual Information (NMI)

Analysis of top central indicators (Tables 2–3) shows that Electric Cars, GDP, Tax Revenue, and Nursing Staff repeatedly ranked among the top five in both degree and betweenness centrality. Their persistent high scores suggest that economic vitality, transport electrification, and health/social service provision are structural connectors in local SDG systems, linking multiple goals over time.

Evolution of the Top 5 Centrality at the Indicator Level (2019-2022)

(a) Degree centrality

Table 2 Top 5 Degree Centrality at the Indicator Level

Year	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5
2019	GDP (0.544)	Tax Rev (0.526)	Car Density (0.509)	Electric Cars (0.509)	Nursing Staff (0.509)
2020	Electric Cars (0.579)	Tax Rev (0.579)	GDP (0.561)	Car Density (0.544)	Local Rec (0.544)

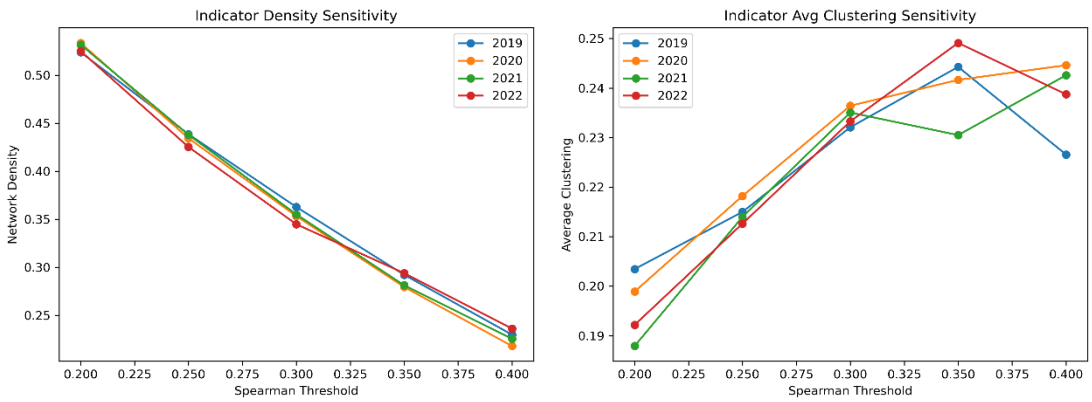
Year	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5
2021	Nursing Staff (0.596)	Local Rec (0.579)	Electric Cars (0.544)	Tax Rev (0.544)	Child Care (0.509)
2022	Nursing Staff (0.579)	Electric Cars (0.561)	Tax Rev (0.544)	GDP (0.526)	Car Density (0.509)

(b) Betweenness Centrality

Table 3 Top 5 Betweenness Centrality at the Indicator Level

Year	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5
2019	GDP (0.857)	Local Rec (0.856)	Electric Cars (0.849)	Prem Mort Men (0.832)	Emp Foreign (0.830)
2020	Electric Cars (0.898)	Prem Mort Men (0.881)	Local Rec (0.878)	GDP (0.874)	Rent Prices (0.852)↑
2021	Electric Cars (0.909)	Prem Mort Men (0.896)	Rent Prices (0.883)↑	GHG Transport (0.878)↑	Gender Inc (0.878)↑
2022	Electric Cars (0.917)	Prem Mort Men (0.906)	Rent Prices (0.894)↑	GHG Transport (0.857)	GDP (0.853)

For sensitivity analysis (Figure 8), Spearman thresholds from 0.20 to 0.40 were tested to evaluate changes in network density and average clustering coefficient. At the indicator level, density decreased monotonically with higher thresholds, while the clustering coefficient peaked at around 0.35 (≈0.24–0.25), indicating optimal modular structure at moderate thresholds. The SDG-level network showed less consistent density trends across years and greater volatility in clustering coefficients, particularly in 2020 and 2021 under high thresholds—indicating sensitivity to edge weight thresholds and potential structural instability.



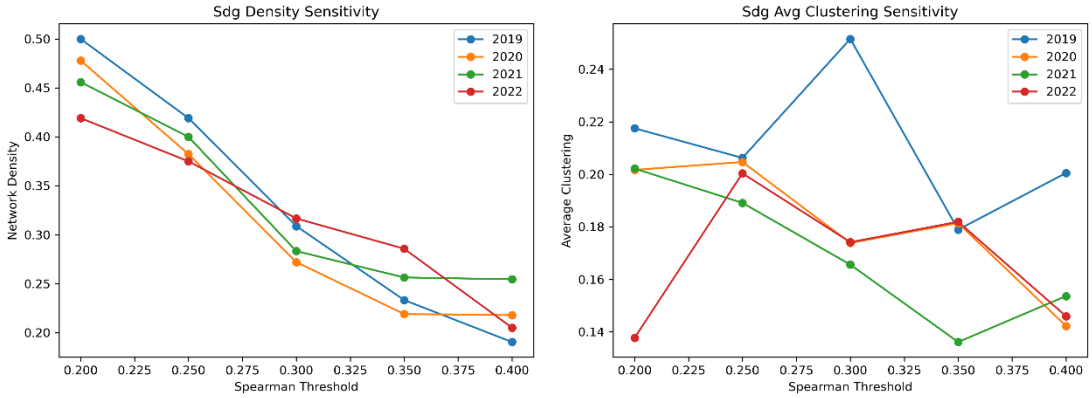


Figure 8 Sensitivity Analysis

To assess robustness under sampling perturbations, bootstrap resampling (n = 100) was applied to the municipality dimension. Aggregated annual networks were reconstructed to estimate confidence intervals for network density and top 5 node centrality. The indicator-level network showed high robustness, with a 95% density interval of [0.358, 0.395] and stable centrality averages (all above 0.60). In contrast, the SDG-level network exhibited more variability, with a density interval of [0.263, 0.343] and broader centrality ranges, suggesting greater sensitivity to sampling noise (Table 4).

Table 4 Network density and statistical estimation of Top K node degree centrality (95% confidence interval)

Level	Index	Mean	95% confidence interval
Indicator	Density (Degree Intensity)	0.377	(0.358, 0.395)
	Top1	0.636	(0.551, 0.722)
	Top2	0.632	(0.546, 0.719)
	Top3	0.624	(0.543, 0.704)
	Top4	0.617	(0.539, 0.695)
	Top5	0.596	(0.552, 0.640)
SDG	Density (Degree Intensity)	0.303	(0.263, 0.343)
	Top1	0.482	(0.400, 0.564)
	Top2	0.456	(0.399, 0.513)
	Top3	0.443	(0.387, 0.500)
	Top4	0.428	(0.362, 0.494)

Level	Index	Mean	95% confidence interval
	Top5	0.412	(0.342, 0.481)

Overall, the indicator-level SDG networks were dynamic but converging toward stability, with recurring key indicators acting as structural hubs. At the SDG level, interactions were more volatile and parameter-sensitive, though evidence of increasing coordination emerged in 2022. These patterns highlight the evolving and partially fragile nature of local SDG interlinkages, suggesting that policy integration is improving but remains uneven across goals and municipalities.

4.4 Community Detection

Using the 2022 municipal-level correlation network, the Louvain algorithm detected four main communities of local SDG indicators (Table 5, Figure 9), with modularity measured at 5.89, indicating a significant underlying community structure characterized by strong internal ties and sparse connections across clusters. Four major communities (with ≥5 nodes) were identified. Their structural properties are as follows:

- Community 1 (19 indicators) combined variables related to urban development (e.g., land use intensity, open space areas), education (school drop-out rates, primary school supply), and employment rates, with an average correlation weight of 0.458 and a mean degree of 6.11. This group suggests systematic co-movement between urban planning, educational resources, and labor market conditions.
- Community 2 (13 indicators) focused on public health (premature mortality, wastewater treatment), social protection (long-term unemployment, SGB II/XII rate), and inequality indicators, exhibiting the highest average weight (0.566), indicating a dense association between health outcomes and socio-economic disadvantages.
- Community 3 (6 indicators) was highly compact, grouping climate-related variables (e.g., GHG emissions from transport and industry, new land use, waste amount), with an average weight of 0.525, reflecting strong interconnections within environmental governance metrics.
- Community 4 (20 indicators) was the most complex cluster, spanning economic performance (GDP, tax revenue, start-ups), urban mobility (electric cars, charging points), and social equity (employment among foreigners, women in councils), showing the highest degree (10.40) and forming a multi-dimensional hub linking multiple SDGs.

Mapping indicators to their respective SDGs highlights the dominant policy domains within each community:

- Community 1 is mainly driven by SDG 11 (Sustainable Cities) and SDG 4 (Education), showing strong synergies between urban services and education access.
- Community 2 links SDG 3 (Good Health), SDG 8 (Work), and SDG 10 (Reduced Inequalities), emphasizing health–equity interdependencies in local governance.
- Community 3 is centered on SDG 13 (Climate Action), indicating that environmental management remains a specialized, self-contained policy sphere.

- Community 4 connects SDG 11 (Cities) and SDG 9 (Industry & Innovation) with social inclusion and environmental indicators, illustrating cross-sector integration in more economically active municipalities.

Table 5 Community structure characteristics and theme distribution

Com	Theme		Node Composition (Full List)	SDG Distribution (Frequency ≥2)
1	Sustainable Cities & Quality Education	19	1. Financial resources balance 2. Watercourse quality 3. Staff in nursing services 4. Employment rate [15-64y] 5. GHG emissions [industry/commerce] 6. Living space 7. Global South partnerships 8. Land use intensity 9. Basic supply [primary school] 10. School drop-out rate 11. Nursing home places 12. Traffic casualties 13. Renewable electricity 14. Fairtrade Schools 15. Child care (<3s) 16. Global South partnerships 17. Open space areas 18. Employment rate [55-64y] 19. Gender employment ratio	SDG 11:3 SDG 4:3 SDG 3:2 SDG 8:2 SDG 17:2
2	Good Health & Decent Work	13	1. Fairtrade Town 2. Long-term unemployment 3. Wastewater treatment 4. Premature mortality [men] 5. Crimes 6. Employed on benefits 7. Liquidity loans 8. Gender income ratio 9. SGB II/XII rate 10. Recreation areas 11. Air pollution 12. GHG emissions [municipal] 13. Premature mortality [women]	SDG 3:3 SDG 8:2 SDG 16:2
3	Climate Action & Sustainable Cities	6	1. New land use 2. Waste amount 3. Floodplain density	SDG 13:3 SDG 11:1

Com	Theme		Node Composition (Full List)	SDG Distribution (Frequency ≥2)
			4. GHG emissions [transport] 5. Naturalisations 6. GHG emissions [industry]	
4	Sustainable Cities & Industry Innovation	20	1. Start-ups 2. Electric cars 3. Rental prices 4. Tax revenue 5. Nature conservation 6. Inclusive childcare 7. Nursing home staff 8. Charging points 9. Broadband supply 10. GDP 11. Drop-out rate [foreigners] 12. Car density 13. Employment [foreigners] 14. High-skilled (%) 15. Women in councils 16. Nitrogen surplus 17. Renewable heating 18. Elderly poverty 19. Land use 20. Drinking water consumption	SDG 11:5 SDG 9:3 SDG 10:2

The network visualization (Figure 9) confirmed this modular structure, with clusters clearly separated and a few bridging nodes (e.g., *Traffic casualties*) acting as potential connectors across communities.

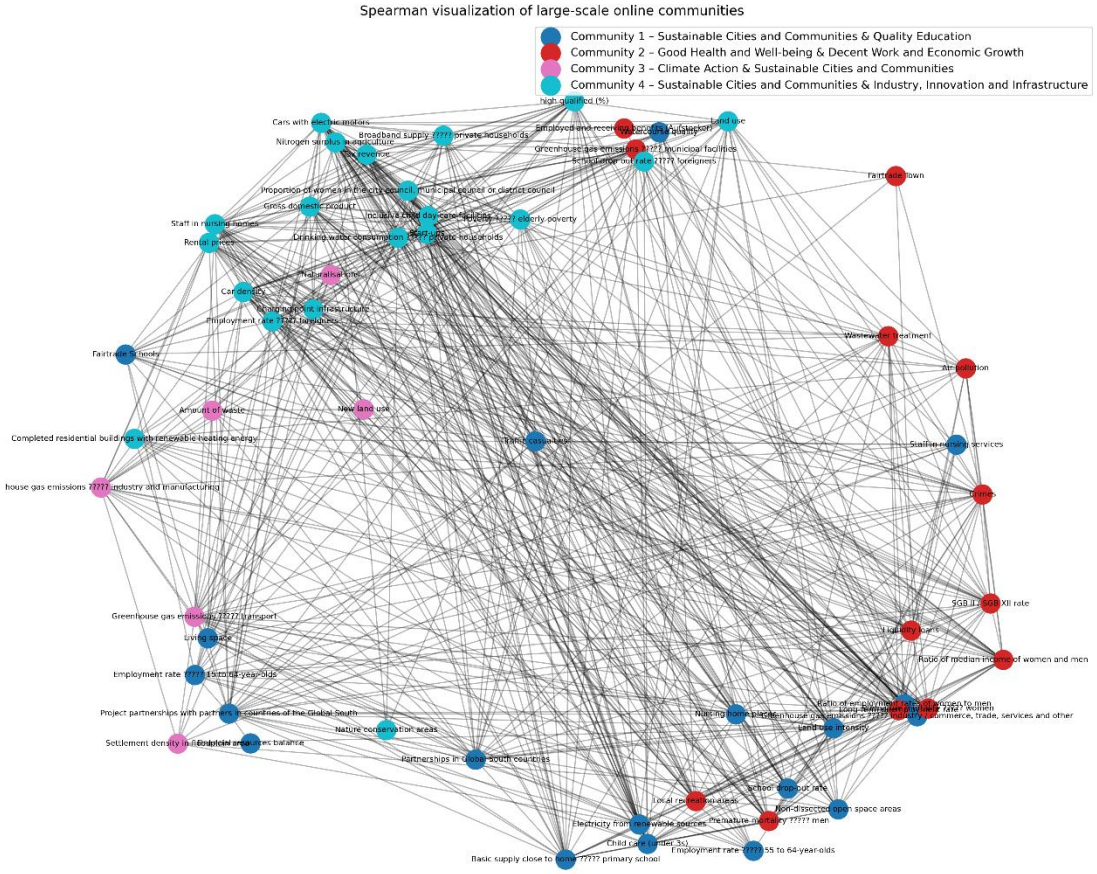


Figure 4 Community Distribution

Community detection results reveal that local SDG systems in Germany exhibit clustered, domain-specific synergies, with urban, educational, and social indicators forming strong ties, while environmental indicators are largely siloed. A subset of economic and infrastructure indicators plays a bridge role, linking otherwise fragmented goal domains. This structure suggests that policy integration is partial and uneven, being stronger for social-urban interactions and weaker for environmental coordination. Similar community structures were identified in the cross-country analysis by Laumann et al. (2020), though our results show a stronger coupling between education and urban goals at the municipal level in Germany.

4.5 Unsupervised Clustering

K-means clustering was applied to the standardized SDG indicator data (2019–2022) to identify regional development patterns among municipalities. The optimal number of clusters was determined as k = 5, based on silhouette scores and the elbow method. The resulting distribution (Figure 10) was unbalanced, with Cluster 1 being the largest (1,198 municipalities) and Cluster 2 the smallest (201 municipalities), indicating that most municipalities share a similar, mid-level SDG profile, while extreme development patterns are less common.

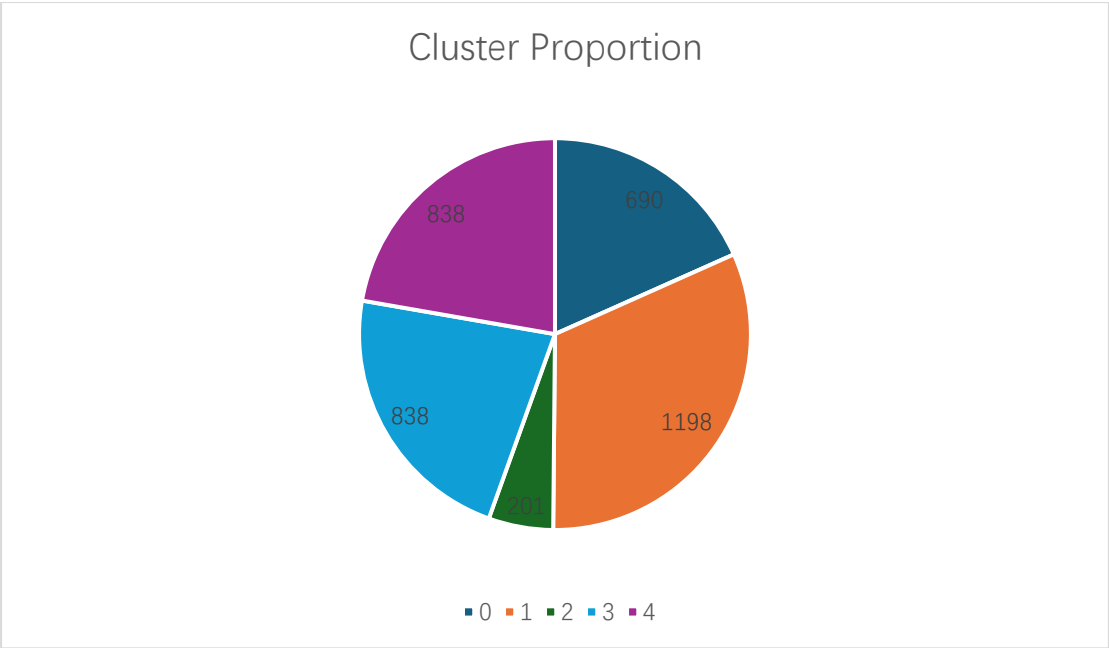


Figure 5 Cluster Proportion

Analysis of cluster centroids (Table 6) highlights distinct sustainability profiles:

- Cluster 0 (690 municipalities) showed negative performance on most goals, especially SDG 6 (Clean Water, –0.86) and SDG 3 (Health, –0.52), representing municipalities with limited resource availability and lower health service outcomes.
- Cluster 1 (1,198 municipalities) displayed balanced but moderate performance across most SDGs, acting as a baseline group typical of many small and medium-sized municipalities.
- Cluster 2 (201 municipalities) performed relatively well in education and gender equality but had low scores on SDG 10 (Reduced Inequalities, –1.15) and SDG 12 (Responsible Consumption, –1.75), indicating social disparities and inefficiencies in resource use despite some strengths in other areas.
- Cluster 3 (393 municipalities) achieved the highest scores on SDG 4 (Education, +1.31), SDG 5 (Gender Equality, +1.60), and SDG 11 (Sustainable Cities, +0.33), suggesting municipalities with strong social inclusion policies and well-developed urban infrastructure.
- Cluster 4 (838 municipalities) was relatively better in SDG 13 (Climate Action, +0.65) and SDG 15 (Life on Land, +0.57) but did not excel in other domains, suggesting environmentally oriented regions with less pronounced progress in social and economic goals.

Table 6 The SDG association strength (Z-score) of each clustering center

SDG	Sustainable Development Goal	Cluster 0	Cluster 1	Cluster 2	Cluster 3	Cluster 4
1	No Poverty	-0.81	**0.66**	-0.38	-0.02	-0.37

SDG	Sustainable Development Goal	Cluster 0	Cluster 1	Cluster 2	Cluster 3	Cluster 4
2	Zero Hunger	-0.09	-0.03	-0.28	-0.21	-0.16
3	Good Health	-0.52	0.28	0.35	**0.84**	-0.46
4	Quality Education	0.14	-0.38	**1.00**	**1.31**	-0.44
5	Gender Equality	-0.21	-0.47	**1.56**	**1.60**	-0.29
6	Clean Water	-0.86	0.52	-0.15	0.20	0.11
7	Affordable Energy	0.15	-0.07	-0.34	0.64	-0.19
8	Decent Work	-0.08	-0.14	0.90	0.34	-0.11
9	Industry Innovation	0.32	-0.002	-0.55	-0.91	0.23
10	Reduced Inequality	**0.71**	-0.12	**-1.15**	-0.58	0.40
11	Sustainable Cities	-0.02	0.01	-0.20	0.33	-0.10
12	Responsible Consumption	0.12	0.28	**-1.75**	-0.08	0.06
13	Climate Action	-0.33	-0.39	**0.90**	-0.51	**0.65**
14	Life Below Water	-0.18	-0.30	-0.31	-0.35	-0.26
15	Life On Land	-0.53	-0.40	-0.08	0.57	**0.57**
16	Peace & Justice	-0.07	0.16	-0.42	-0.31	0.06
17	Partnerships	-0.28	-0.29	-0.29	-0.29	-0.27

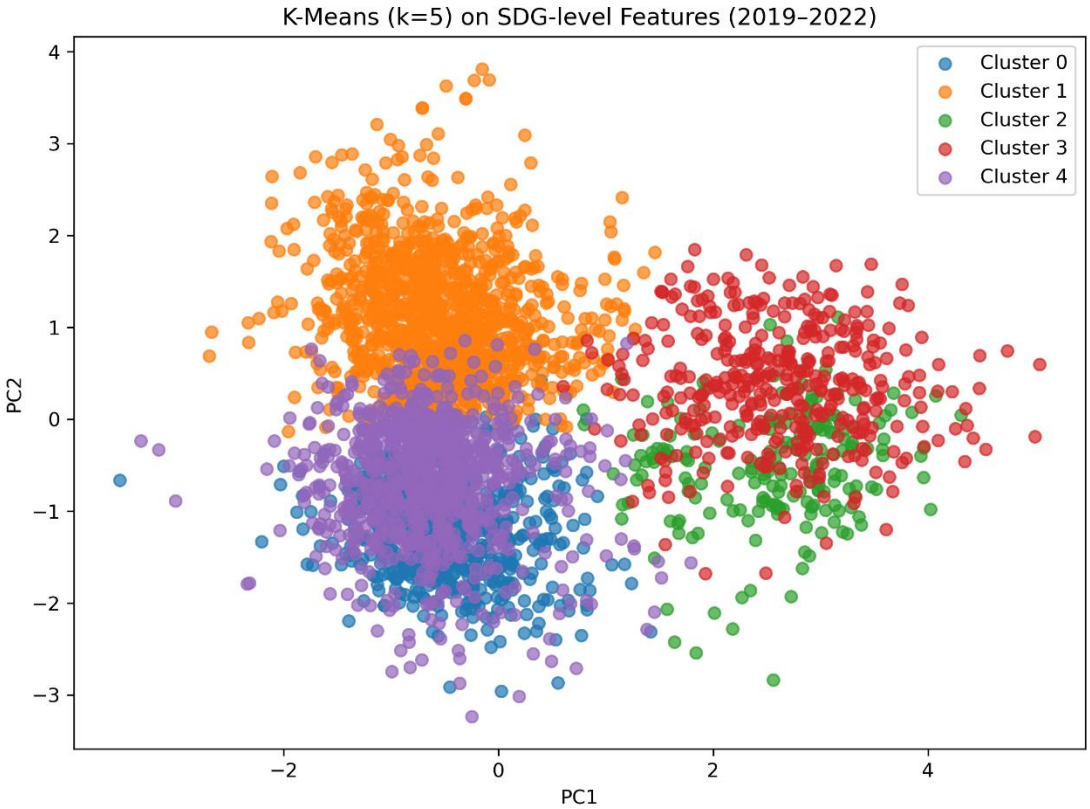


Figure 11 PCA Distribution

PCA visualization (Figure 11) confirmed clear separability between clusters, with Clusters 2 and 3 positioned furthest apart, reflecting contrasting performances in social equity and responsible consumption.

The clustering results underline heterogeneity in local SDG performance across Germany. Certain municipalities excel in social and educational goals but struggle with inequality and consumption patterns, while others show stronger environmental focus at the expense of broader development. These distinct profiles imply that policy priorities and resource allocation are highly region-specific, requiring tailored strategies rather than uniform SDG interventions.

4.6 Supervised Regression

To examine directional relationships among SDGs, a multi-output random forest regression was implemented, treating each SDG as a dependent variable in turn while using the remaining 16 goals as predictors. This approach identifies the relative influence of other goals on each target and allows for near-term forecasting of SDG trajectories at the municipal level.

Regression performance varied substantially across SDGs (Table 7).

- High predictability: Sustainable Cities (SDG 11) showed the lowest mean squared error (0.038 ± 0.002), indicating that urban development outcomes are highly dependent on other goals, particularly education, infrastructure, and governance factors. Zero Hunger (SDG 2) and Decent Work (SDG 8) also exhibited relatively strong predictability (MSE < 0.1), implying a clear structural linkage to broader socio-economic conditions.
- Low predictability: Climate Action (SDG 13) and Life on Land (SDG 15) had high errors

(MSE > 0.6), suggesting weak or non-linear connections with other goals, likely reflecting localized environmental factors not fully captured by socio-economic predictors.

Overall, the results reinforce findings from earlier correlation analyses, where social and economic goals formed a tightly coupled system, while environmental goals remained more isolated.

Table 7 Random Forest Multi-Target Regression Performance with Top-5 Predictive Features

SDG	MSE (Mean ± Std)	Top 1 Feature	Top 2	Top 3	Top 4	Top 5
1 (Poverty)	0.196 ± 0.009	3 (Health)	16 (Peace)	4 (Education)	6 (Water)	9 (Industry)
2 (Hunger)	0.052 ± 0.012	6 (Water)	12 (Consumption)	10 (Inequality)	5 (Gender)	1 (Poverty)
3 (Health)	0.115 ± 0.008	10 (Inequality)	1 (Poverty)	4 (Education)	9 (Industry)	11 (Cities)
4 (Education)	0.207 ± 0.016	5 (Gender)	9 (Industry)	3 (Health)	1 (Poverty)	11 (Cities)
5 (Gender)	0.229 ± 0.040	4 (Education)	3 (Health)	8 (Work)	10 (Inequality)	6 (Water)
6 (Water)	0.239 ± 0.016	1 (Poverty)	15 (Land)	16 (Peace)	11 (Cities)	10 (Inequality)
7 (Energy)	0.339 ± 0.054	11 (Cities)	4 (Education)	16 (Peace)	1 (Poverty)	13 (Climate)
8 (Work)	0.084 ± 0.007	5 (Gender)	12 (Consumption)	4 (Education)	11 (Cities)	16 (Peace)
9 (Industry)	0.187 ± 0.014	4 (Education)	3 (Health)	16 (Peace)	5 (Gender)	10 (Inequality)
10 (Inequality)	0.204 ± 0.013	3 (Health)	5 (Gender)	1 (Poverty)	11 (Cities)	9 (Industry)
11 (Cities)	0.038 ± 0.002	9 (Industry)	3 (Health)	4 (Education)	12 (Consumption)	7 (Energy)
12 (Consumption)	0.239 ± 0.053	8 (Work)	6 (Water)	10 (Inequality)	11 (Cities)	5 (Gender)
13 (Climate)	0.669 ± 0.076	11 (Cities)	7 (Energy)	3 (Health)	5 (Gender)	8 (Work)
14 (Below water)	0.125 ± 0.019	10 (Inequality)	15 (Land)	3 (Health)	5 (Gender)	1 (Poverty)
15 (Land)	0.629 ± 0.061	5 (Gender)	11 (Cities)	3 (Health)	13 (Climate)	8 (Work)
16 (Peace)	0.183 ± 0.081	9 (Industry)	6 (Water)	5 (Gender)	1 (Poverty)	11 (Cities)
17 (Partnerships)	0.061 ± 0.037	11 (Cities)	10 (Inequality)	15 (Land)	13 (Climate)	9 (Industry)

Feature importance scores (Figure 12) highlight several consistent relationships:

- Education (SDG 4) is the most influential predictor for Gender Equality (SDG 5) and Reduced Inequalities (SDG 10), with normalized importance values exceeding 0.40. This underscores the role of educational access and quality as a key enabler of inclusive development.
- Urban development (SDG 11) repeatedly predicts outcomes for Energy (SDG 7), Climate Action (SDG 13), and Partnerships (SDG 17), suggesting that improvements in housing, infrastructure, and mobility strongly shape progress in other domains.
- Health (SDG 3) and Peace, Justice & Institutions (SDG 16) are important drivers for No Poverty (SDG 1), aligning with previous evidence that governance quality and healthcare provision are fundamental for poverty alleviation.
- Conversely, environmental goals rarely act as strong predictors, indicating their limited influence on socio-economic outcomes within the current municipal data structure.

These patterns suggest that local SDG systems are predominantly driven by human capital, governance, and spatial planning, while ecological outcomes are less interconnected and potentially more dependent on external or long-term factors.

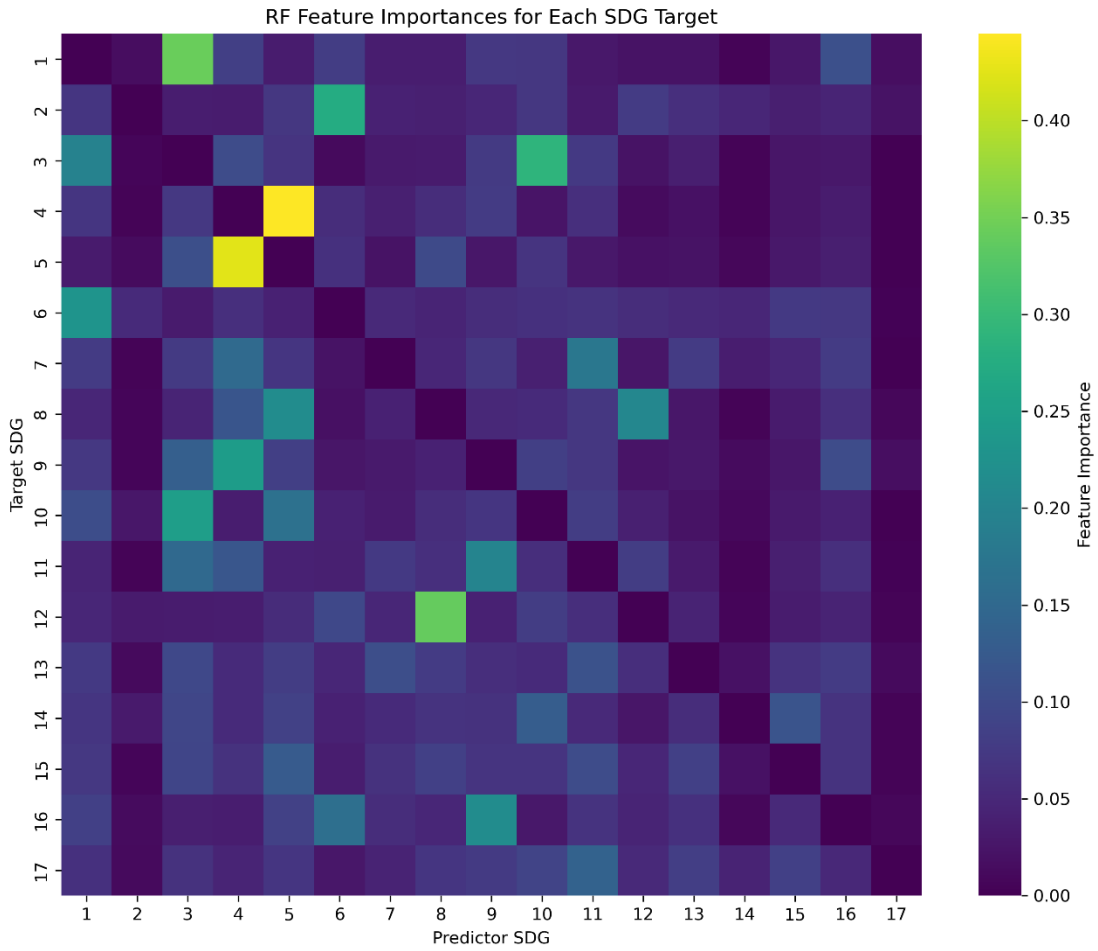


Figure 12 Random Forest Feature Importances for Each SDG Target

Using fitted regression models, linear projections were made for 2023–2025 (Table 8). Key findings include:

- Positive outlook: Education (SDG 4) is expected to increase by ~17% annually, Industry & Innovation (SDG 9) by ~15%, and Good Health (SDG 3) by ~2%, pointing to continued progress in social infrastructure and technological development.
- Stagnation: Goals such as Water (SDG 6), Sustainable Cities (SDG 11), Climate (SDG 13), Peace & Justice (SDG 16), and Partnerships (SDG 17) show minimal changes (<2%), suggesting either near-saturation or limited policy innovation in these domains.
- Negative trends: Decent Work (SDG 8) is forecast to decline sharply, reaching –515.8 by 2025, which may indicate rising unemployment or structural shifts in labor markets. Responsible Consumption (SDG 12) is also projected to worsen, falling to –1.75, highlighting ongoing challenges in sustainable production and consumption patterns.

These forecasts imply that future local SDG progress will remain uneven, with strong advances in education and innovation, moderate improvements in health, but potential setbacks in labor market sustainability and consumption efficiency.

Table 8 Linear Forecasts of SDG Indicators (2023-2025)

SDG	Sustainable Development Goal	2023	2024	2025	Annual Change Rate*
1	No Poverty	5.21	5.27	5.33	+1.1%
2	Zero Hunger	27.57	18.09	8.60	-43.5%
3	Good Health	42.53	43.46	44.39	+2.2%
4	Quality Education	268.85	317.85	366.86	+16.8%
5	Gender Equality	93.44	95.55	97.65	+2.2%
6	Clean Water	91.40	91.38	91.37	-0.01%
7	Affordable Energy	1.14	1.11	1.08	-2.6%
8	Decent Work	124.76	-195.53	-515.81	N/A
9	Industry Innovation	49.57	57.40	65.24	+14.7%
10	Reduced Inequality	100.06	96.34	92.61	-3.8%
11	Sustainable Cities	140.95	140.61	140.27	-0.2%
12	Responsible Consumption	4.22	-12.69	-29.60	N/A
13	Climate Action	9.45	9.52	9.59	+0.7%
14	Life Below Water	30.87	30.87	30.87	0%
15	Life On Land	15.30	14.81	14.31	-3.3%

SDG	Sustainable Development Goal	2023	2024	2025	Annual Change Rate*
16	Peace & Justice	553.38	562.62	571.85	+1.6%
17	Partnerships	0.031	0.031	0.031	-0.2%

To identify municipalities with unusual SDG performance, an unsupervised anomaly detection method, the Isolation Forest—was applied. A total of 166 outlier municipalities were detected (see Appendix). This method is suitable for high-dimensional data and can uncover municipalities that deviate significantly from the general SDG pattern, potentially due to policy bias, data anomalies, or unique local development paths. These anomalies serve as valuable reference points for subsequent case studies and policy calibration.

The supervised regression results emphasize three main points. The first one is the Central role of education and urban development. These goals not only show strong predictive power but also consistently act as upstream enablers for equity, governance, and environmental outcomes. This supports the hypothesis of social foundations driving broader SDG progress, consistent with findings from Horvath et al. (2022) and Vinuesa et al. (2019). The second one is environmental goals are weakly embedded. Despite their importance, goals like SDG 13 and 15 neither predict nor are well-predicted by other SDGs. This could reflect policy siloing, long time lags for environmental interventions, or data quality limitations at the local level. The third one is future risks in employment and consumption patterns. The negative projections for SDG 8 and SDG 12 suggest that economic growth is not translating into sustainable labor and resource outcomes, pointing to possible trade-offs within local development strategies.

The strong predictive role of education for gender equality and reduced inequalities aligns with the global findings of Asadikia et al. (2021), while our results additionally highlight its linkage to sustainable cities at the local level. For policy implication, municipalities aiming for integrated sustainability should prioritize cross-cutting investments in education, governance quality, and spatial planning, while urgently addressing the disconnect between economic growth, labor market stability, and sustainable consumption. Strengthening data-driven coordination between environmental and social domains could improve predictability and long-term alignment of SDG outcomes.

4.7 Causal Analysis

To move beyond static correlations and explore directional relationships among SDG indicators, a constraint-based structure learning algorithm was combined with time-lagged fixed-effects regression. The analysis uncovered 20 statistically significant causal links (Table 9, $p < 0.05$), revealing how socio-economic, environmental, and infrastructural factors interact dynamically within local sustainability systems.

The most influential pathways highlight social equity and economic vitality as upstream drivers of sustainability transitions. For example, improvements in the Gender Income Ratio consistently led to higher Electric Vehicle Adoption in the subsequent year ($\beta = 4.90$, $R^2 = 0.81$), suggesting that gender-inclusive labor markets may facilitate access to clean mobility technologies, potentially by expanding purchasing power across population groups. Similarly,

GDP growth positively influenced both EV adoption ($\beta = 3.17$, $R^2 = 0.80$) and Rental Prices ($\beta = 1.92$, $R^2 = 0.77$), indicating that local economic expansion fuels both sustainable transport uptake and housing market activity. These findings emphasize that social and economic policies indirectly shape environmental outcomes, forming cascading effects across SDGs.

Land use indicators also emerged as significant determinants of future development patterns. Land Use Intensity had a strong negative causal effect on New Land Development ($\beta = -12.25$, $p = 1.7\times10^{-4}$), confirming a form of path dependence where municipalities with historically dense land use face physical or regulatory constraints in expanding further. Similarly, increases in Local Recreation Areas were associated with reduced Undeveloped Open Space, indicating trade-offs between urban amenities and natural land preservation.

Other causal chains linked resource consumption and environmental pressures. Drinking Water Consumption predicted Nitrogen Surplus in agriculture ($\beta = 2.74$, $p = 1.5\times10^{-5}$), showing how household-level resource use can drive downstream ecological impacts. Socio-economic vulnerability also exerted environmental influence: a rise in Social Welfare Recipients predicted Open Space Loss ($\beta = 1.61$), possibly reflecting pressures from low-cost housing developments or urban sprawl in disadvantaged areas.

Table 9 Top 20 Lagged Effect Relationships with Statistical Significance

Rank	Cause Variable (X)	Effect Variable (Y)	β Coefficient	p-value	R^2	Domain
1	Gender Income Ratio	Electric Car Adoption	+4.90	$<1\times10^{-10}$	0.813	Social-Economic
2	GDP	Electric Car Adoption	+3.17	$<1\times10^{-10}$	0.804	Economic
3	Land Use	New Land Development	-12.25	1.70×10^{-4}	0.197	Environmental
4	Gender Income Ratio	Rental Prices	+2.93	$<1\times10^{-10}$	0.753	Social-Economic
5	Nursing Home Staff	Electric Car Adoption	+2.69	$<1\times10^{-10}$	0.504	Social
6	Drinking Water Consumption	Agricultural Nitrogen Surplus	+2.74	1.54×10^{-5}	0.024	Environmental
7	Land Use Intensity	New Land Development	-3.65	4.11×10^{-3}	0.068	Environmental
8	Local Recreation Areas	Undeveloped Open Space	-3.05	5.09×10^{-9}	0.036	Environmental
9	Nursing Services Staff	Electric Car Adoption	-2.64	$<1\times10^{-10}$	0.506	Social
10	Car Density	Electric Car Adoption	+2.33	2.22×10^{-16}	0.215	Transportation

Rank	Cause Variable (X)	Effect Variable (Y)	β Coefficient	p-value	R^2	Domain
11	GDP	Rental Prices	+1.92	$<1\times10^{-10}$	0.766	Economic
12	Crimes	Electric Car Adoption	-1.84	1.18×10^{-3}	0.225	Social
13	Nursing Home Places	Electric Car Adoption	-1.79	1.11×10^{-3}	0.032	Social
14	Rental Prices	Electric Car Adoption	+1.77	$<1\times10^{-10}$	0.663	Economic
15	Local Recreation Areas	New Land Development	-1.75	3.93×10^{-2}	0.015	Environmental
16	Agricultural Nitrogen Surplus	Electric Car Adoption	-1.74	$<1\times10^{-10}$	0.789	Environmental
17	Local Recreation Areas	Renewable Electricity	-1.71	2.27×10^{-4}	0.032	Energy
18	Social Welfare Recipients	Undeveloped Open Space	+1.61	2.79×10^{-10}	0.020	Social
19	Tax Revenue	Financial Resources Balance	-1.61	2.02×10^{-2}	0.093	Economic
20	Drinking Water Consumption	Gender Income Ratio	-2.39	4.28×10^{-4}	0.026	Social

Network-level analysis (Figure 13) classified indicators into High out-degree "drivers" (Table 10) and High in-degree "receivers" (Table 11). Gender Income Ratio (7.60), GDP (5.31), Nursing Home Staff (4.81), and Car Density (3.81) were top systemic influencers. These indicators shape multiple other outcomes, highlighting that equity, economic strength, and social services function as leverage points in local SDG systems. Electric Vehicle Adoption (8.89), Renewable Electricity (4.70), and Rental Prices (2.62) were the most causally sensitive outcomes, depending on a wide set of prior social, economic, and environmental conditions. This asymmetry shows that social structure and economic activity drive long-term sustainability transitions, while technology adoption, energy use, and housing markets are primarily response variables.

A. Top 15 Driver Indicators (High Out-Degree)

Table 10 Top 15 Driver Indicators (High Out-Degree)

Rank	Indicator	Out-Degree	Domain	Key Relationships
1	Gender Income Ratio	7.60	Social-Economic	→ EVs, Rentals, GDP
2	GDP	5.31	Economic	→ EVs, Rentals
3	Nursing Home Staff	4.81	Social	→ EVs, Healthcare
4	Social Welfare Rate	4.26	Social	→ Open Space
5	Car Density	3.81	Transportation	→ EV Adoption
6	Long-term Unemployment	3.64	Economic	→ Social Services
7	High-Skilled Workforce (%)	3.29	Labor	→ Innovation
8	Rental Prices	2.35	Economic	→ EVs, Housing
9	Elderly Poverty Rate	1.18	Social	→ Social Spending
10	Charging Infrastructure	0.77	Energy	→ EV Adoption

B. Top 15 Responsive Indicators (High In-Degree)

Table 11 Top 15 Responsive Indicators (High In-Degree)

Rank	Indicator	In-Degree	Domain	Main Drivers
1	Electric Vehicle Adoption	8.89	Transportation	Gender, GDP, Nursing
2	Renewable Electricity	4.70	Energy	Land Use, Policy
3	Naturalisations	4.13	Social	Economy, Crime
4	Rental Prices	2.62	Economic	GDP, Income

Rank	Indicator	In-Degree	Domain	Main Drivers
5	Primary School Access	2.49	Education	Poverty, Planning
6	Watercourse Quality	1.57	Environmental	Agriculture, Runoff
7	Charging Infrastructure	1.31	Energy	Policy, EV Demand
8	GDP	1.17	Economic	Trade, Investment
9	Crime Rates	1.04	Social	Unemployment, Policing
10	Nursing Home Staff	0.98	Social	Aging, Health Policy

The resulting causal network (Figure 13) exhibited dense interconnections among socio-economic nodes, with multiple arrows converging on technological and environmental outcomes. Few feedback loops were detected, suggesting that most identified causal effects are unidirectional, from social/economic determinants to sustainability impacts. This highlights a potential leverage hierarchy: Equity and education → Economic vitality → Infrastructure and technology adoption → Environmental outcomes.

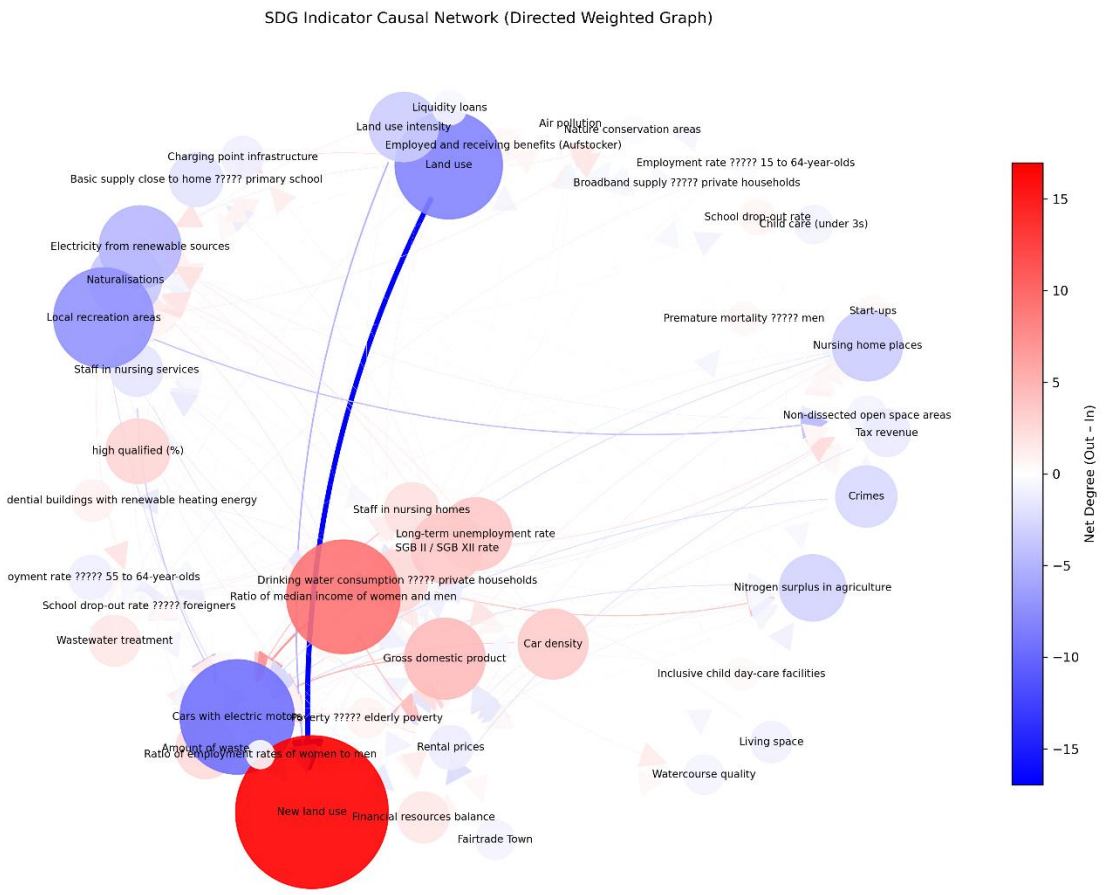


Figure 6 SDG Indicator Causal Network

Overall, the causal analysis depicts a hierarchical system of local SDG dynamics, where equity, education, and economic activity drive subsequent changes in infrastructure, technology, and environmental indicators. Feedback loops are rare, indicating largely one-way causal flows. These findings imply that isolated environmental interventions may have limited impact unless accompanied by broader social and economic reforms, particularly those enhancing equality and governance quality. Strengthening cross-domain policies targeting foundational drivers could thus yield more systemic and enduring progress toward multiple SDGs.

5. Conclusion and Future Work

5.1 Conclusion

This study set out to tackle a long-standing gap in sustainable development research: the lack of systematic, data-driven evidence on how Sustainable Development Goals (SDGs) interact within local governance contexts. Although the 2030 Agenda highlights that progress on one goal often depends on its connections with others, most empirical studies have remained at the national or global level. These studies tend to rely on expert judgments or highly aggregated indicators, which inevitably obscure the complexities of local realities. By drawing on subnational SDG data from German municipalities and combining them with machine learning techniques, this research has attempted to fill that gap. In doing so, it provides not only new empirical insights but also a methodological basis for more integrated, evidence-informed local sustainability planning.

The results show that SDG interactions at the municipal scale are neither random nor fixed. Instead, they form evolving networks in which certain goals consistently play more central roles. In particular, indicators linked to education, gender equality, and sustainable cities emerged as structural hubs that connect multiple other domains. This finding suggests that investments in areas such as social equity and urban infrastructure may generate wider spillover effects, reaching into domains like clean energy adoption, labor market performance, and even housing affordability. On the other hand, environmental goals—including those related to climate action and land use—appeared more peripheral and less tightly connected. This implies that ecological priorities are often pursued in relative isolation from social and economic agendas. Such dynamics point to the importance of viewing local sustainability strategies as part of an interconnected system, in which targeted interventions can have wider, and sometimes unexpected, consequences.

Beyond these structural patterns, the analysis also highlighted strong regional differences. The clustering results revealed that municipalities do not follow a single development trajectory. Some appear to move forward in a balanced way across both social and environmental dimensions, while others continue to lag behind on key issues such as water management, innovation capacity, or inequality reduction. These differences make it clear that a one-size-fits-all national strategy is unlikely to succeed. Municipalities instead need differentiated approaches—tailored to their own socio-economic and environmental conditions but also informed by comparisons with similar cities facing comparable challenges. Recognizing such heterogeneity is crucial if regional disparities in SDG progress are to be reduced and more equitable forms of sustainable development achieved.

The analysis also identified specific indicators that act as leverage points within the SDG system. Variables such as gender income equality, urban mobility infrastructure, and economic productivity consistently influenced a range of other sustainability outcomes, from electric vehicle adoption rates to rental market dynamics. These findings suggest that interventions in these domains can trigger cascading effects throughout the local SDG network, accelerating progress across multiple targets simultaneously. At the same time, lagged regression analysis revealed path-dependent relationships, such as the long-term constraints imposed by historical land use intensity on new urban expansion. These results highlight that sustainable development is not only multi-dimensional but also temporally structured, where past decisions set trajectories that shape future possibilities. Policies designed without

acknowledging such dependencies risk locking municipalities into unsustainable paths that are difficult to reverse.

From a theoretical perspective, this research advances the literature on SDG interlinkages by demonstrating that local-scale networks can be empirically constructed and that they exhibit distinct structural and temporal dynamics compared to national-level analyses. It shows that social and environmental transitions are deeply intertwined, with social equity often serving as a precondition for environmental innovation, rather than merely a parallel goal. Methodologically, the study offers a novel combination of network analysis, clustering, supervised regression, anomaly detection, and causal inference, integrated into a single analytical framework. This approach moves beyond static, pairwise correlations, providing a more dynamic and interpretable understanding of how sustainable development unfolds in complex, decentralized systems. It illustrates how machine learning can complement traditional econometric approaches, offering more granular, data-driven tools for decision support while retaining interpretability.

Practically, the results of this research have significant implications for local policy. They suggest that municipalities should move away from treating SDGs as isolated targets and instead design strategies that recognize systemic interdependencies. Prioritizing education, gender equity, and urban livability can amplify positive effects across the SDG spectrum, accelerating overall progress. At the same time, understanding heterogeneity among municipalities allows for more precise allocation of resources and the development of peer-learning networks where leading cities mentor those struggling with similar challenges. Detecting anomalous municipalities with atypical SDG profiles provides an opportunity for early policy intervention, whether to address data quality issues, correct misaligned priorities, or respond to emerging sustainability risks before they become entrenched.

Although this research has focused on Germany, the methodological framework developed here is highly transferable to other contexts. Many countries face similar challenges of fragmented local data, lack of structured interlinkage modeling, and absence of causal evidence for policy prioritization. Applying this approach elsewhere could enable cross-country benchmarking, support the design of regionally adapted sustainability strategies, and ultimately strengthen the global evidence base on how SDGs interact in practice. By moving local SDG monitoring beyond descriptive dashboards to predictive, prescriptive, and system-based analytics, this study contributes to the development of more coherent, synergistic, and resilient pathways towards achieving the 2030 Agenda.

Nevertheless, the study is not without limitations. The availability and quality of municipal SDG data remain uneven, which may have influenced network structure estimates and causal inferences. Although lagged regression offers suggestive evidence of directional effects, it cannot fully eliminate confounding factors or establish causality with the same rigor as quasi-experimental designs. The analysis also treated municipalities as independent units, overlooking spatial spillovers such as commuting flows or pollution diffusion that likely influence SDG dynamics. Furthermore, while machine learning enhanced analytical power, translating its results into transparent, actionable guidance for policymakers remains an ongoing challenge. Future research should refine these methods, integrate higher-frequency and alternative data sources, and explore spatial and temporal dependencies more explicitly, ensuring that data-driven approaches are both scientifically robust and practically relevant.

In conclusion, this study demonstrates that machine learning can fundamentally transform the way local governments understand and manage sustainable development. By uncovering the network structures, leverage points, and causal mechanisms that shape SDG progress, it provides a foundation for more integrated, evidence-based decision-making at the municipal level. The findings highlight that advancing sustainability is not merely about achieving individual targets but about navigating a complex web of interdependencies, where strategic, well-informed interventions can generate far-reaching benefits. The framework developed here offers both a methodological contribution to academic research and a practical toolkit for policymakers, supporting the goal of coherent, synergistic progress towards a more sustainable future.

5.2 Future Work

While this study represents a step forward in understanding local SDG interlinkages through a novel machine learning framework, several important avenues remain open for future research. These avenues stem from both the methodological limitations identified in this study and the broader ambition to develop more robust, predictive, and policy-oriented models of sustainable development at the municipal level.

One key priority is to address data availability and quality constraints. The municipal SDG dataset employed in this research, though valuable, contains a substantial proportion of missing observations, particularly in years prior to 2019. Among the current 70 indicators, approximately 30% still lack stable, consistently reported data across municipalities. This limitation constrains the accuracy of network construction and causal inference. Future work should therefore focus on filling these data gaps through complementary approaches, including targeted field surveys, systematic data mining from local government reports, and the integration of alternative data sources such as satellite observations, sensor networks, or even text-based information from social media and policy documents. Moreover, it is essential to periodically evaluate the feasibility and representativeness of the selected indicators, ensuring that the SDG dataset dynamically evolves as new information becomes available and outdated metrics are replaced with more relevant ones.

Beyond improving data completeness, there is a need to strengthen causal inference methodologies in future analyses. The current study relied on time-lagged panel regressions and structure learning to identify directional dependencies among indicators. These approaches cannot fully account for unobserved confounders or policy endogeneity. Future research could incorporate quasi-experimental designs, such as difference-in-differences, synthetic control methods, or instrumental variable strategies, to more rigorously assess how specific local policies—for example, climate adaptation measures, public transport expansion, or education reforms—alter SDG interlinkages over time. Such approaches would move the field from mapping potential relationships to establishing credible evidence on cause-and-effect mechanisms.

The spatial dimension of SDG interactions also deserves greater attention. Municipalities are embedded in broader regional systems where economic flows, migration patterns, and environmental externalities frequently cross administrative boundaries. Ignoring these spatial spillovers risks misrepresenting the true nature of SDG interdependencies. Future work could employ spatial econometric models or leverage emerging techniques such as graph neural networks to capture these cross-border influences, enabling the design of coordinated

regional sustainability strategies rather than fragmented, municipality-specific plans.

Finally, there is an urgent need to make complex machine learning outputs more accessible to policymakers. In this study, diagnostics were produced on network structures, leverage points, and unusual municipal profiles, but these outputs remain quite technical and not immediately usable. A natural next step for future research would be to experiment with more interactive tools—such as decision-support platforms or scenario simulators—that can translate analytical results into clearer guidance. Developing explainable AI models could also be important, since policymakers are more likely to engage with methods they can understand and trust. Such efforts would give local authorities the chance to explore different policy options, anticipate possible unintended effects, and design more integrated strategies that are both locally relevant and consistent with the broader 2030 Agenda. Looking ahead, several directions appear especially promising. Closing remaining data gaps, strengthening causal inference, and incorporating spatial dependencies are all important for advancing the scientific study of SDG interlinkages. At the same time, more attention should be given to the usability of research outputs, so that findings are not only rigorous but also practical for those working in local government. Taken together, these improvements would create a stronger and more adaptive analytical foundation for municipalities striving to promote sustainable development in increasingly complex contexts.

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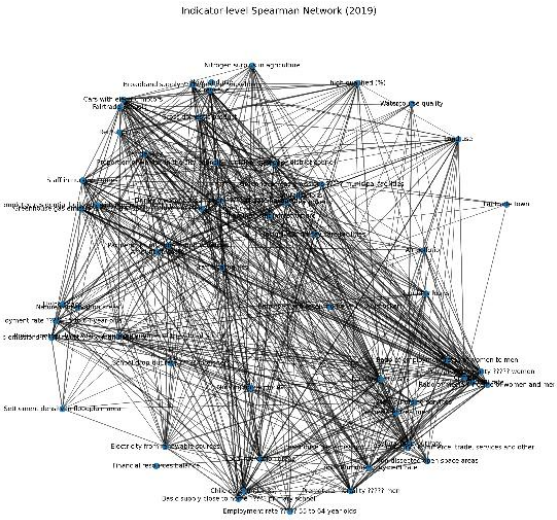
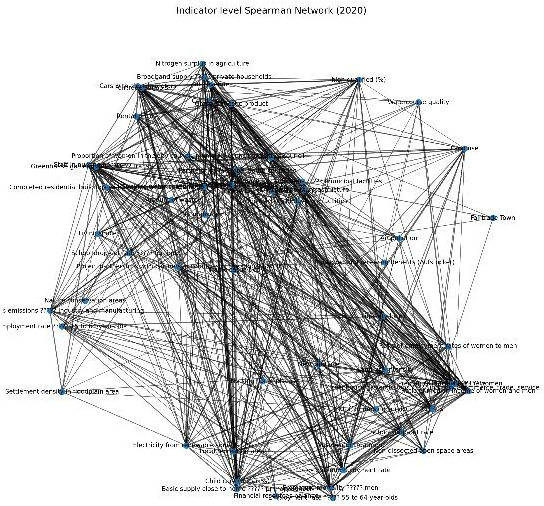
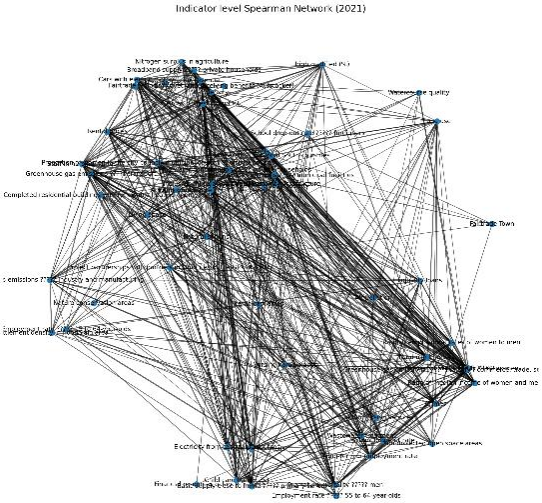
Appendix

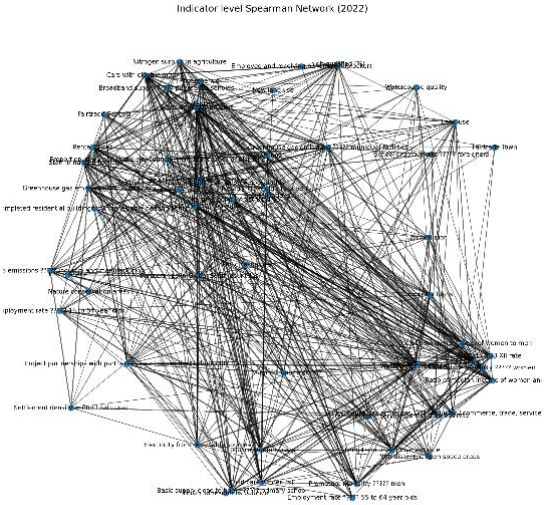
The code and output are in this link
https://drive.google.com/drive/folders/10dmiTBOTTL1noccMghfweJZHBxUlGeEW?usp=sharing

70 indicators spearman network:

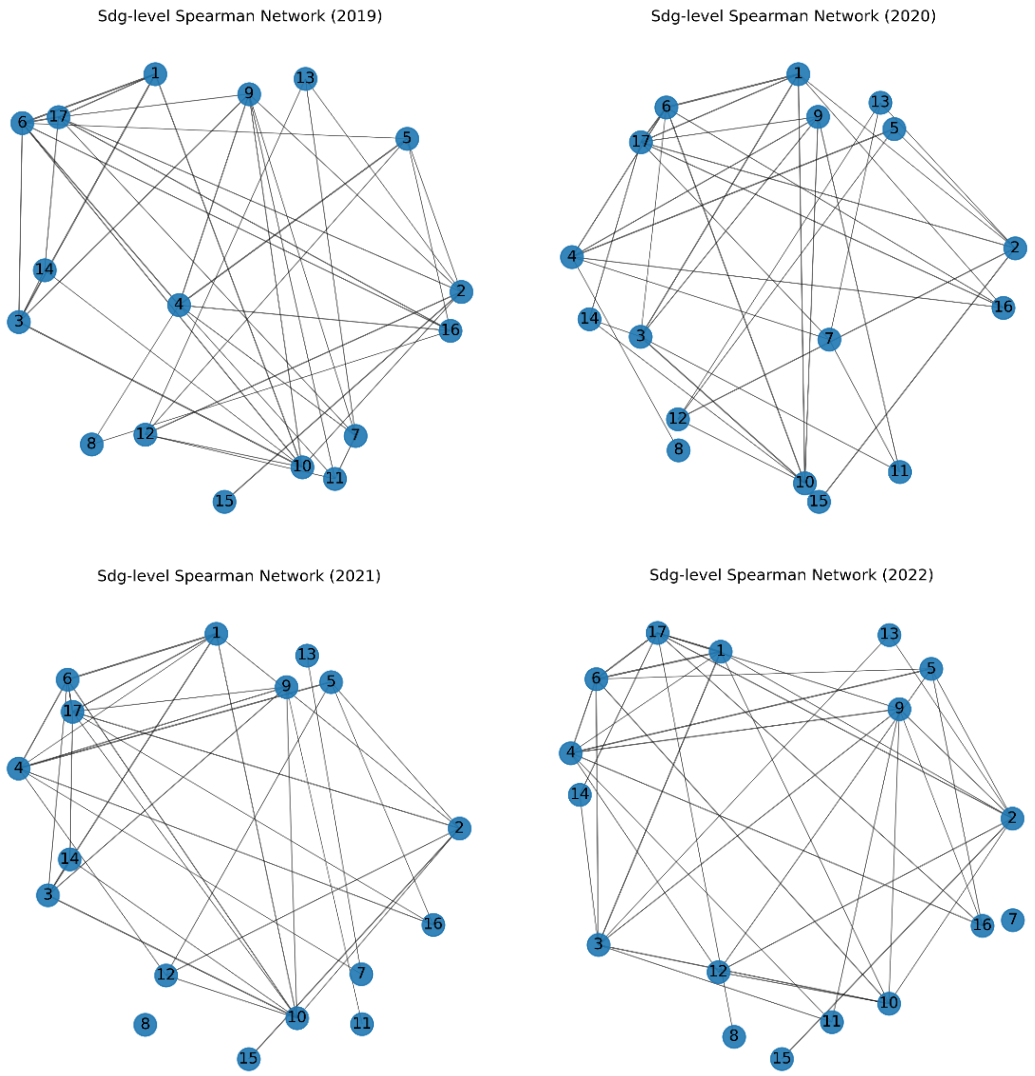








17 SDG goals spearman network:



166 outlier municipalities:

['Altenberg, Stadt', 'Amt Neuhaus', 'Annaburg, Stadt', 'Arneburg-Goldbeck', 'Aschheim', 'Aue-Bad Schlema, Stadt', 'B??rdeland', 'B??tzow, Stadt', 'Bad F??ssing', 'Bad Hindelang, M', 'Bad Schmiedeberg, Stadt', 'Bad Wiessee', 'Barnim, Landkreis', 'Beetzendorf-Diesdorf', 'Belgern-Schildau, Stadt', 'Berchtesgaden, M', 'Berlin', 'Berlin, Stadt', 'Bischofswiesen', 'Bismark (Altmark), Stadt', 'Bobritzsch-Hilbersdorf', 'Br??hl', 'Bremen', 'Bremen, Stadt', 'Bremerhaven, Stadt', 'Chemnitz, Stadt', 'Coswig (Anhalt), Stadt', 'Crimmitschau, Stadt', 'D??sseldorf, Stadt', 'Dietzenbach, Kreisstadt', 'Dillingen/ Saar, Stadt', 'Dingolfing, St', 'Dithmarschen, Kreis', 'Dohna, Stadt', 'Drebach', 'Dresden, Stadt', 'Ebersbach-Neugersdorf, Stadt', 'Egelner Mulde', 'Eifelkreis Bitburg-Pr??m, Landkreis', 'Elbe-Havel-Land', 'Elbe-Heide', 'Elbe-Parey', 'Erzgebirgskreis, Landkreis', 'Essen, Stadt', 'Falkenberg/Elster, Stadt', 'Falkenstein/Harz, Stadt', 'Fl??ha, Stadt', 'Flechtingen', 'Floh-Seligenthal', 'Frankfurt am Main, Stadt', 'Freital, Stadt', 'Garching b.M??nchen, St', 'Garmisch-Partenkirchen, M', 'Gelsenkirchen, Stadt',

'Geratal', 'Glash??tte, Stadt', 'Gmund a.Tegernsee', 'Goldene Aue', 'Gommern, Stadt', 'Gr??ditz, Stadt', 'Gr??felfing', 'Gr??nwald', 'Hagen, Stadt der FernUniversit??t', 'Hamburg', 'Hamburg, Freie und Hansestadt', 'Harzgerode, Stadt', 'Heidenau, Stadt', 'Hellenthal', 'Herne, Stadt', 'Herrnhut, Stadt', 'Herten, Stadt', 'Holzkirchen, M', 'Idar-Oberstein, Stadt', 'Ismaning', 'Jerichow, Stadt', 'Jerichower Land, Landkreis', 'K??ln, Stadt', 'Kaiserslautern, Stadt', 'Kalbe (Milde), Stadt', 'Karst??dt (Landkreis Prignitz)', 'Kiefersfelden', 'Kirchberg, Stadt', 'Klingenthal, Stadt', 'L????nitz, Stadt', 'L??wenberger Land', 'Leverkusen, Stadt', 'Luckau, Stadt', 'M??ckern, Stadt', 'M??lsen', 'Mainz, Stadt', 'Mansfelder Grund-Helbra', 'Marburg, Universit??tsstadt', 'Marienberg, Stadt', 'Mecklenburgische Seenplatte, Landkreis', 'Meerane, Stadt', 'Meinerzhagen, Stadt', 'Merzig-Wadern, Landkreis', 'Mittelsachsen, Landkreis', 'Mittenwald, M', 'Monheim am Rhein, Stadt', 'Murnau a.Staffelsee, M', 'Neubiberg', 'Neubrandenburg, Stadt', 'Neuffen, Stadt', 'Neukirchen/Erzgeb.', 'Neustadt in Sachsen, Stadt', 'Niederg??rsdorf', 'Nohfelden', 'Nuthe-Urstromtal', 'Oberaudorf', 'Oberding', 'Oberhausen, Stadt', 'Oberkochen, Stadt', 'Oberlungwitz, Stadt', 'Oberstdorf, M', 'Oelsnitz/Erzgeb., Stadt', 'Oranienbaum-W??rlitz, Stadt', 'Osterburg (Altmark), Hansestadt', 'Osternienburger Land', 'Panketal', 'Perl', 'Pl??n, Stadt', 'Planegg', 'Prignitz, Landkreis', 'Pullach i.Isartal', 'Recklinghausen, Kreis', 'Regionalverband Saarbr??cken, Regionalverband', 'Rochlitz, Stadt', 'Rottach-Egern', 'Ruhpolding', 'S??chsische Schweiz-Osterzgebirge, Landkreis', 'S??deifel', 'Saale-Wipper', 'Saarbr??cken, Landeshauptstadt', 'Satow', 'Sch??nau a.K??nigssee', 'Sch??ningen, Stadt', 'Schalksm??hle', 'Schipkau', 'Schwarzenberg/Erzgeb., Stadt', 'Sebnitz, Stadt', 'Seegebiet Mansfelder Land', 'Seehausen (Altmark)', 'Sehmatal', 'Tangerh??tte, Stadt', 'Teltow-Fl??ming, Landkreis', 'Thalheim/Erzgeb., Stadt', 'Tharandt, Stadt', 'Trier, Stadt', 'Trier-Saarburg, Landkreis', 'Uckermark, Landkreis', 'Unterf??hring', 'Velburg, St', 'Walldorf, Stadt', 'Wei??wasser/O.L., Stadt / B????a Woda', 'Weida-Land', 'Werdau, Stadt', 'Westliche B??rde', 'Wethautal', 'Wiesbaden, Landeshauptstadt', 'Wolfsburg, Stadt', 'Wuppertal, Stadt', 'Wusterhausen/Dosse', 'Zeithain', 'Zw??nitz, Stadt', 'Zwickau, Stadt']

Statement of Risk Assessment & Ethics Approval Requirements
Student Candidate Number [NWNK1]: Student Name: [E[redacted] Student UCL Email Address: [zc[redacted] **Supervisor Name:** [Manos Chaniotakis]: Supervisor UCL Email Address: [m.chaniotakis@ucl.ac.uk]: **Dissertation Research Proposal**: • Title / Topic: Identification of Synergies and Causal Relationships among Sustainable Development Goals at the Local Level in Germany: An Empirical Study Based on Machine Learning. • Research Question(s) / Aims & Objectives: This work aims at utilising advanced language models and ML techniques to extract interlinkages from data available on a local level. • Data & source (specify all data to be used; if none, explain why): The datasets are given by the supervisor. • Method(s) (specify all methods to be used): machine learning, network analysis.
I have read and understood **Step A1 'Does the research require a Risk Assessment?'** and: [DELETE ONE STATEMENT]: • This planned research does NOT require a risk assessment.
I have read and understood **Step A2 'Does the research require External research ethics approval?'** and: [DELETE ONE STATEMENT]: • This planned research does NOT require external ethics review.
External ethics approval is *not required* and I have read and understood **Step A3 'Is the research Exempt from the need for ethics approval?'** and: [DELETE ONE STATEMENT]: • This planned research IS EXEMPT from the need for research ethics approval.
The research is *not exempt* from the need for ethics approval and I have read and understood **Step A4 'Does the research require High Risk ethics approval?'** and: [DELETE ONE STATEMENT]: • This planned research is NOT deemed high risk.
The research is *not exempt* from the need for ethics approval, does not require high risk ethics approval and: I have read and understood **Step A5 'Does the research require ESDA low risk ethics review for questions-based methods OR BSEER low risk ethics review for other methods?'** and: [DELETE ONE STATEMENT]: • This planned research requires MSc ESDA Low Risk Ethics approval for questions-based methods and approval will be secured before data collection starts.
I confirm that: • the information I have provided is accurate to the best of my knowledge. • if the answers to any of these questions changes, I will go through this protocol again.